

Digital Images and Symbolic Encoding of Guatemalan Polyphonic Choirbooks

Enhancing Preservation and Access for Early Music Sources through

Digitization and Music Information Retrieval

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PhD Music Technology (McGill University)

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Universidade Federal de Rio de Janeiro, Brazil

December 11th, 2024



Do mini, in nomine Domine,
E ne dictus quive nit Benedictus
quive nit Benedictus quivenit in nomine Do
mini, in nomine Do mini, in nomine Do
mini, in nomine Do mini,

Some Background

I am from Guatemala

I was fortunate to do my Master's and PhD studies in
Music Technology in Canada (at McGill University)

And what I wanted with this project was to **take all that music technology background** I got during my studies and **apply it to the musical heritage of my home country (Guatemala)** to help in its preservation

Introduction

Motivation and Corpus

Introduction: Motivation

- Digitization and encoding of music books from Guatemala
- Part of Guatemala's colonial past and an important part of its cultural heritage
- **Goal:** Preserve and enhance access to this music
- Through the digitization and music information retrieval (MIR) technologies

Introduction: Corpus

- Six choirbooks held at the Archivo Histórico Arquidiocesano de Guatemala (AHAG), an archive located next to the Metropolitan Cathedral in Guatemala City
- Manuscripts (handwritten music)
- Average page dimensions: 30 cm x 45 cm
- Material: Paper
- Copied in the 17th and 18th centuries
- Contain mostly Renaissance European polyphonic music (16th c.), which continued to be used in Guatemala until the 19th c.

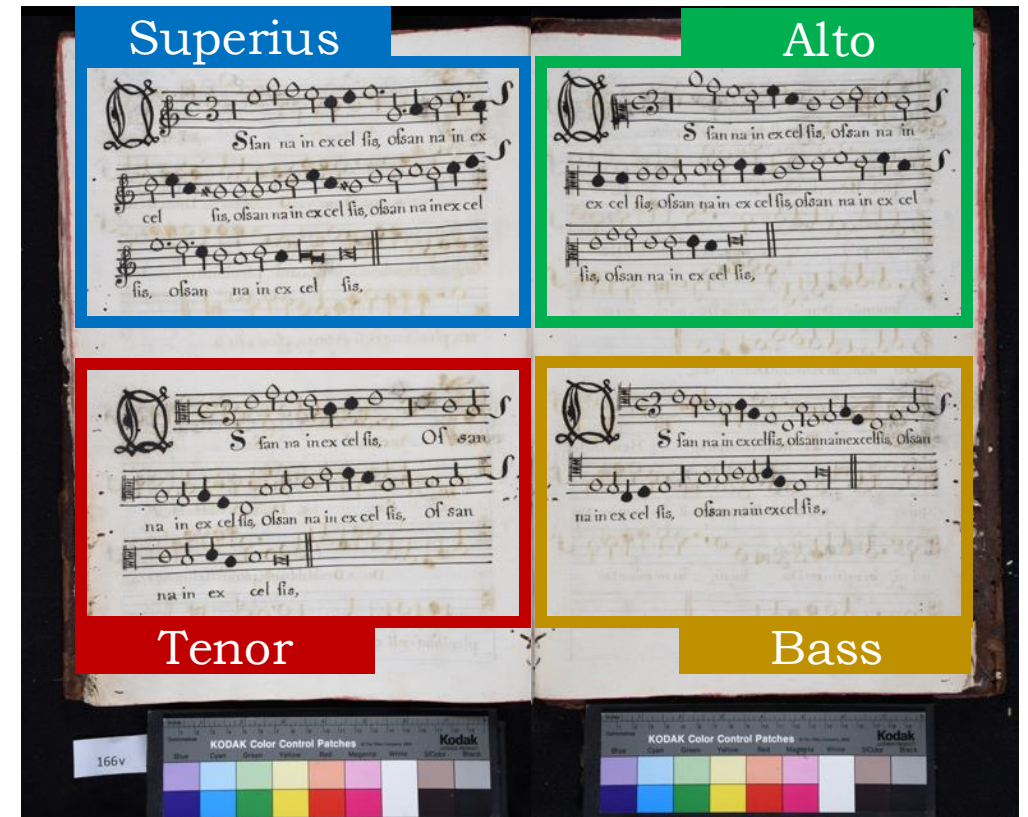
Introduction: Corpus

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Introduction: Corpus

Contain mostly Renaissance European **polyphonic music** (16th c.)

- Each of the different voices (soprano, alto, tenor, bass) sing their own melody
- Rather than the whole choir singing the same melody (**monophonic** music)
- Written in **mensural notation**
- Written in a **separate-parts layout** (**voices separated** rather than lined up in a **score layout**)



Introduction: Corpus (*Mensural Music Books*)

- Usually in some **separate-parts** layout



Score

Superius

S fan na in ex cel fis, ofsan na in ex cel fis, ofsan na in ex cel fis, ofsan na in ex cel fis.

Alto

S fan na in ex cel fis, ofsan na in ex cel fis, ofsan na in ex cel fis, ofsan na in ex cel fis.

S fan na in ex cel fis, Of san na in ex cel fis, Ofsan na in ex cel fis, Ofsan na in ex cel fis.

Tenor

S fan na in ex cel fis, ofsan na in ex cel fis, ofsan na in ex cel fis, ofsan na in ex cel fis.

Bass

KODAK Color Control Patches

166v

KODAK Color Control Patches

The image displays a musical score for a four-part setting of "Osanna in excelsis". The parts are arranged vertically: Superius (top), Altus, Tenor, and Bassus (bottom). Each part is represented by a staff with a clef and a key signature of one flat (B-flat). The Superius and Altus parts begin with a whole rest, while the Tenor and Bassus parts begin with a half note. The lyrics "Os - san-na - in - ex - cel - sis" are written below the staves, with hyphens indicating the syllables are spread across multiple notes. The Superius and Altus parts end with a half note, while the Tenor and Bassus parts end with a half note. The Bassus part includes a repeat sign at the end of the phrase.

which implies dealing with the issue of this early music notation (the interpretation of the notes' duration)

You need an expert to interpret the duration of the notes in this notation

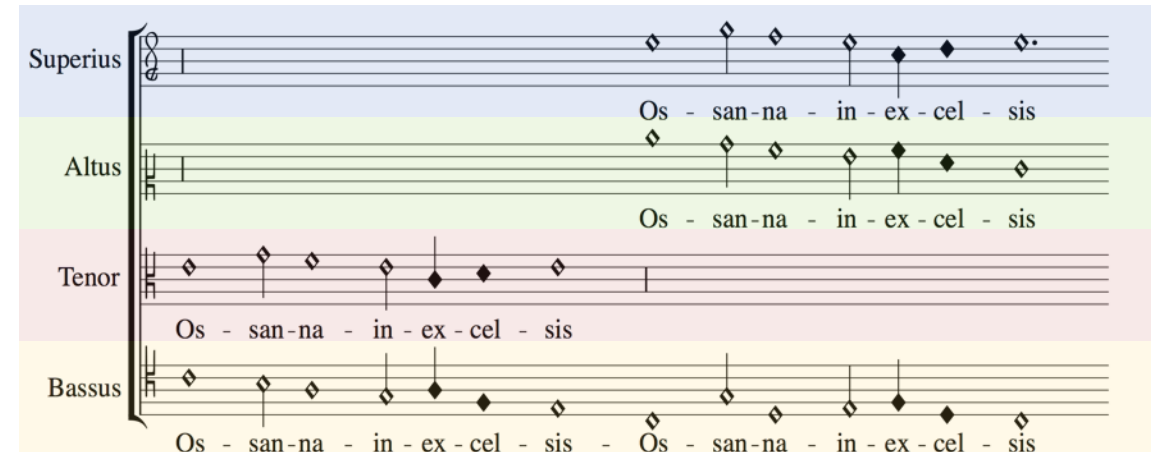
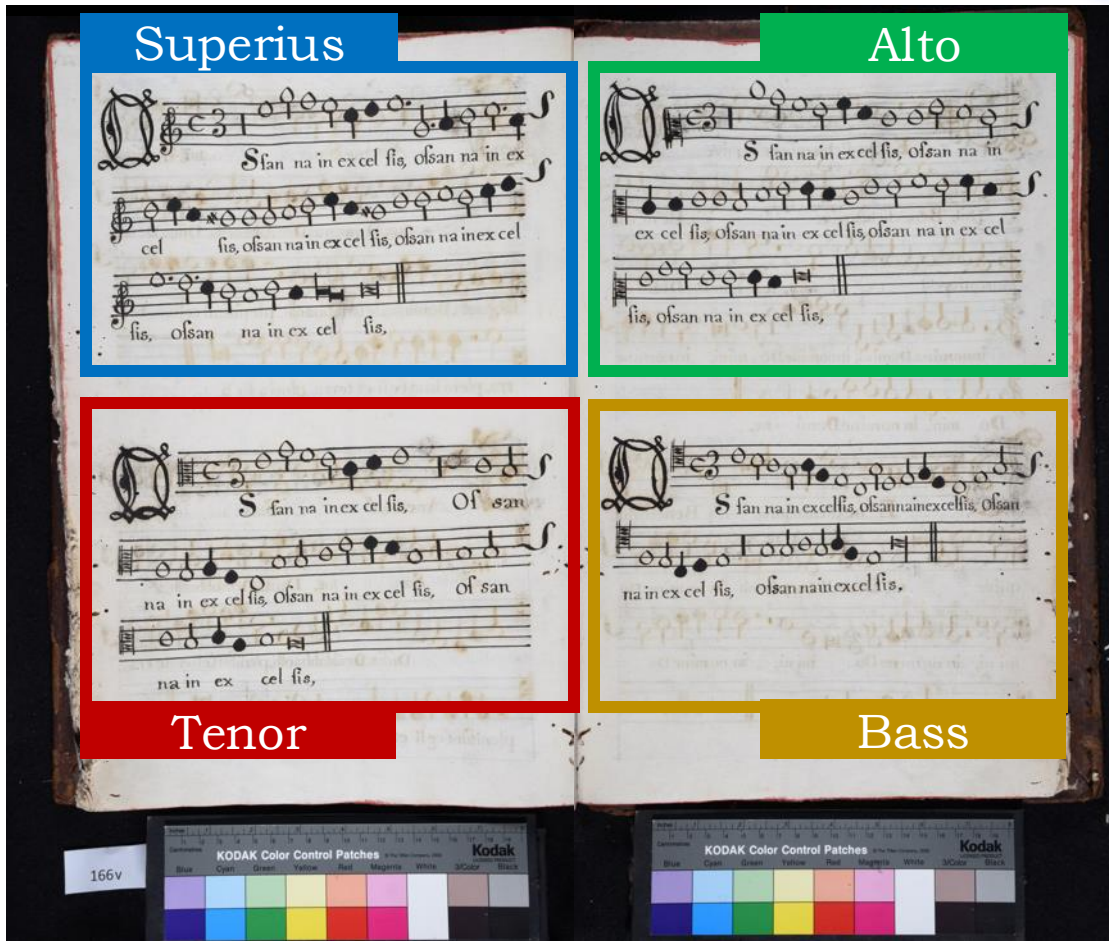
Introduction: Corpus (*Mensural Music Books*)

- Usually in some **separate-parts** layout



Symbolic Score

- A score encoded in a *symbolic* or *machine-readable* format



which implies dealing with the issue of this early music notation (the interpretation of the notes' duration)

You need an expert to interpret the duration of the notes in this notation

Introduction: Motivation

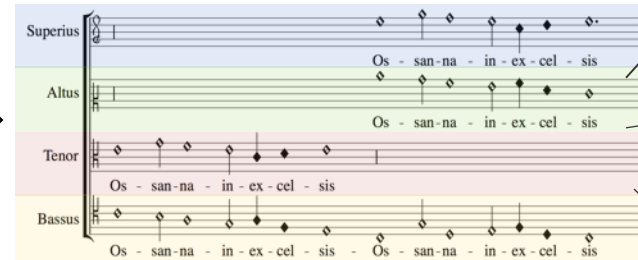
- **Goal:** Preserve and enhance access to this music
- Through the digitization and music information retrieval (MIR) technologies
- **Pilot project:** First book



Manuscript



Digital images
of the mensural pieces



Symbolic Score
interpreted durations,
an issue in mensural notation

playback

automatic
transcription into
modern values

comparison to
concordant sources

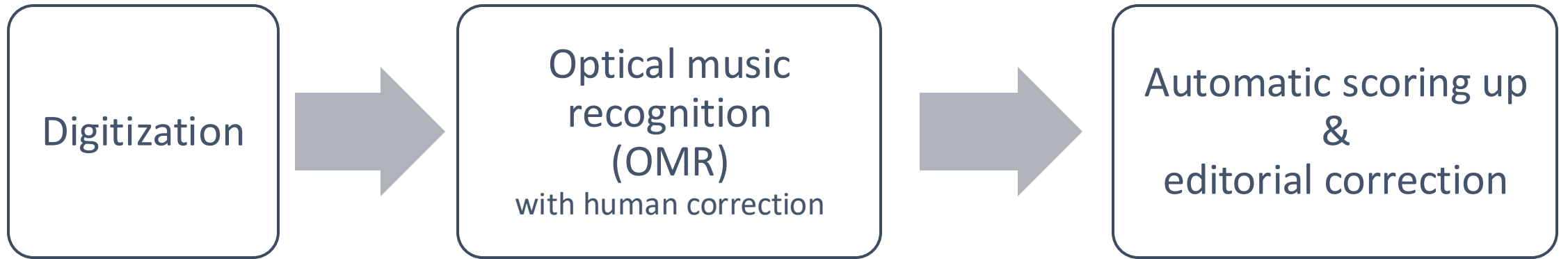
Introduction: Motivation

- **Goal:** Preserve and enhance access to this music
- Through the **digitization** and **music information retrieval (MIR)** technologies:
 - A **do-it-yourself (DIY) book scanner** for high-resolution images
 - **Optical music recognition (OMR)** software trained for handwritten mensural notation
 - An **interpreter for mensural notation**
 - A **music-analysis tool** served as an **error checker**
- Present these tools and their integration into a **digitization and MIR pipeline** to create the digital images and symbolic scores

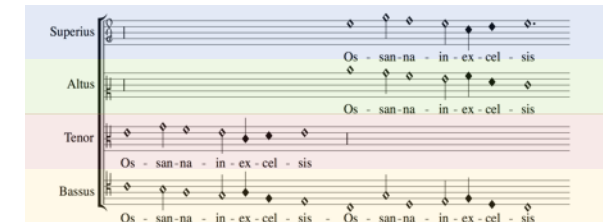
Work

The Digitization and Music
Information Retrieval (MIR) Pipeline

Digitization & MIR Pipeline

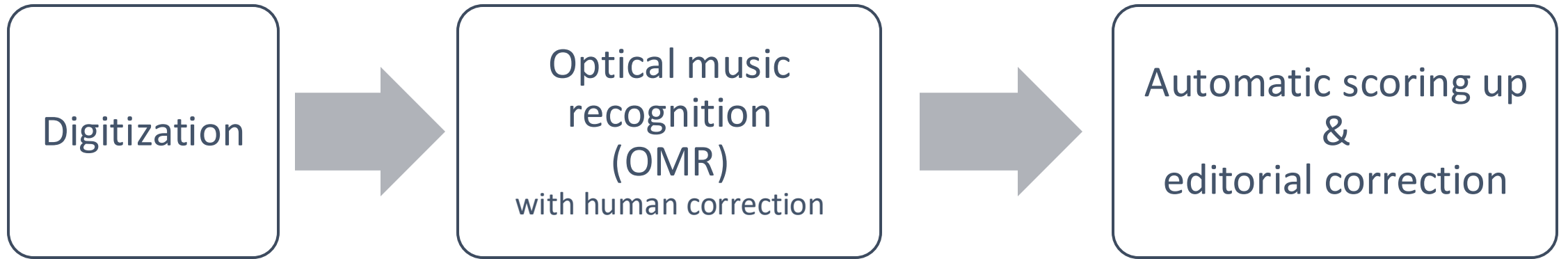


Manuscript



Score file
interpreted durations

Digitization & MIR Pipeline



Manuscript

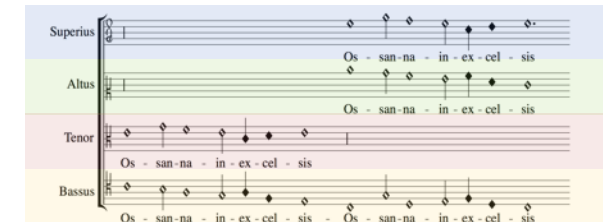


Digital images

```
<mei>
<music>
<staff n="1">
<clef line="2" shape="G"/>
<note dur="semibrevis" pname="g" oct="5"/>
<note dur="semibrevis" pname="d" oct="5"/>
<dot/>
<note dur="minima" pname="c" oct="5"/>
<ligature form="recta">
<note dur="semibrevis" pname="d" oct="5"/>
<note dur="semibrevis" pname="g" oct="4"/>
</ligature>
<note dur="brevis" pname="b" oct="4"/>
</staff>
</music>
</mei>
```

Parts file

encodes the **symbols** in each
part/voice of the images



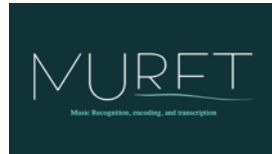
Score file
interpreted durations

Digitization & MIR Pipeline

Digitization



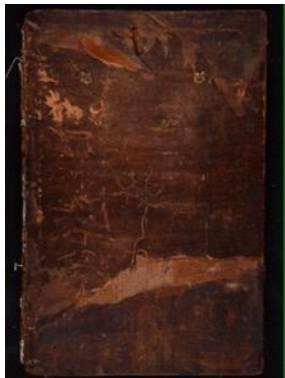
Optical music recognition (OMR) with human correction



Automatic scoring up & editorial correction

MEASURING POLYPHONY

DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC



Manuscript

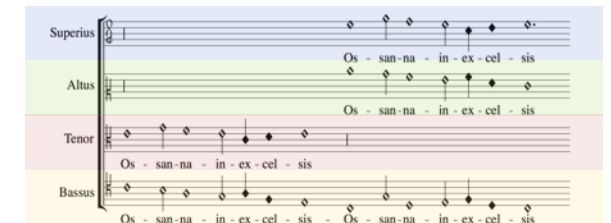


Digital images

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<mei>
<music>
<staff n="1">
  <clef line="2" shape="G"/>
  <note dur="semibrevis" pname="g" oct="5"/>
  <note dur="semibrevis" pname="d" oct="5"/>
  <dot/>
  <note dur="minima" pname="c" oct="5"/>
  <ligature form="recta">
    <note dur="semibrevis" pname="d" oct="5"/>
    <note dur="semibrevis" pname="g" oct="4"/>
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  <note dur="brevis" pname="b" oct="4"/>
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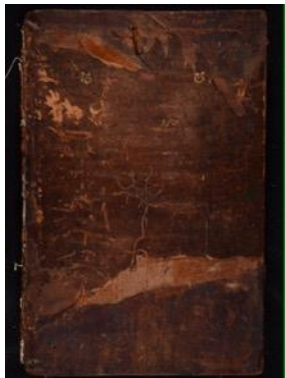
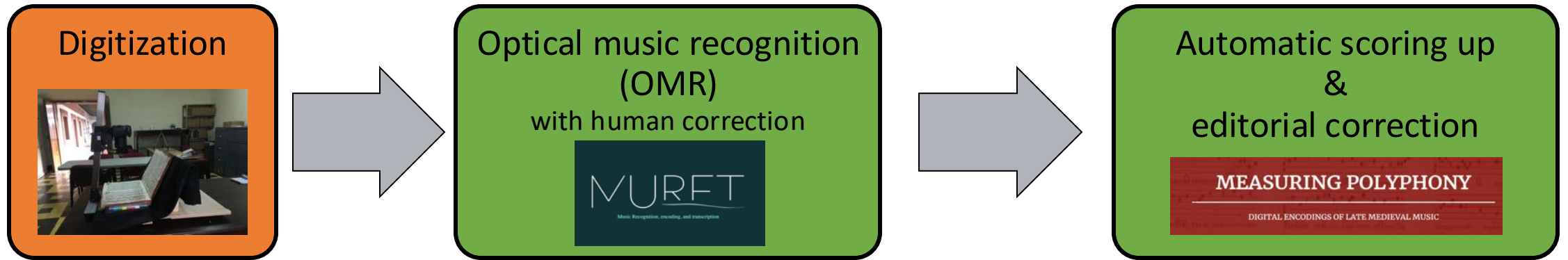
Parts file

encodes the **symbols** in each
part/voice of the images



Score file
interpreted durations

Digitization & MIR Pipeline



Manuscript



Digital images

```
<mei>
<music>
<staff n="1">
  <clef line="2" shape="G"/>
  <note dur="semibrevis" pname="g" oct="5"/>
  <note dur="semibrevis" pname="d" oct="5"/>
  <dot/>
  <note dur="minima" pname="c" oct="5"/>
  <ligature form="recta">
    <note dur="semibrevis" pname="d" oct="5"/>
    <note dur="semibrevis" pname="g" oct="4"/>
  </ligature>
  <note dur="brevis" pname="b" oct="4"/>
</staff>
</music>
</mei>
```

Parts file

encodes the **symbols** in each
part/voice of the images



Score file
interpreted durations

1. Digitization

Obtaining digital images

Do-it-yourself book scanner

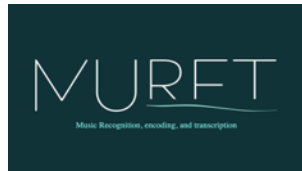
Digitization

Do-it-yourself (DIY) book scanner

Digitization



Optical music recognition
(OMR)



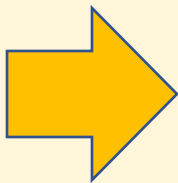
Automatic scoring up
&
editorial correction

MEASURING POLYPHONY

DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC



Manuscript

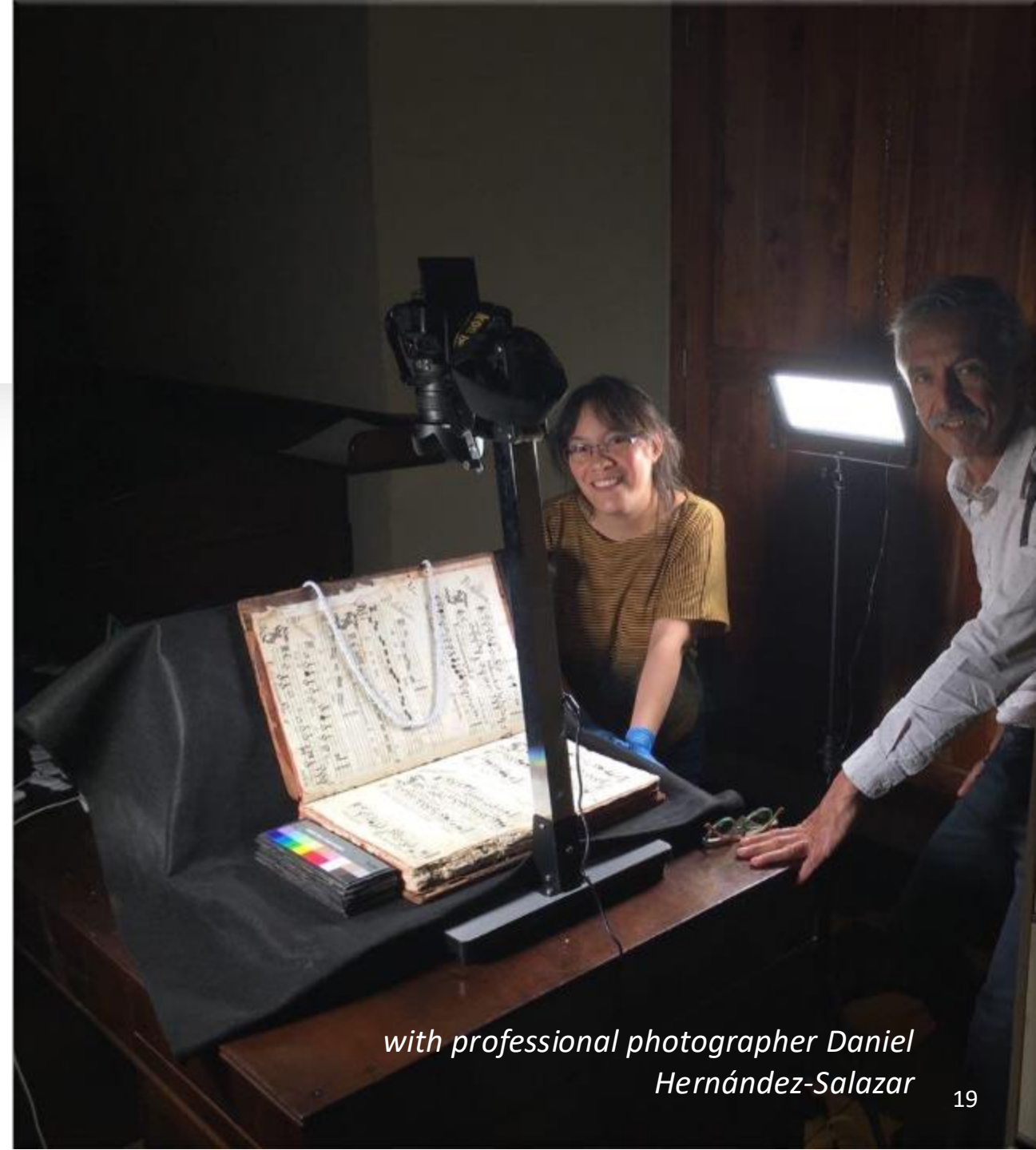


Digital image

Technology Used

DIY Book Scanner

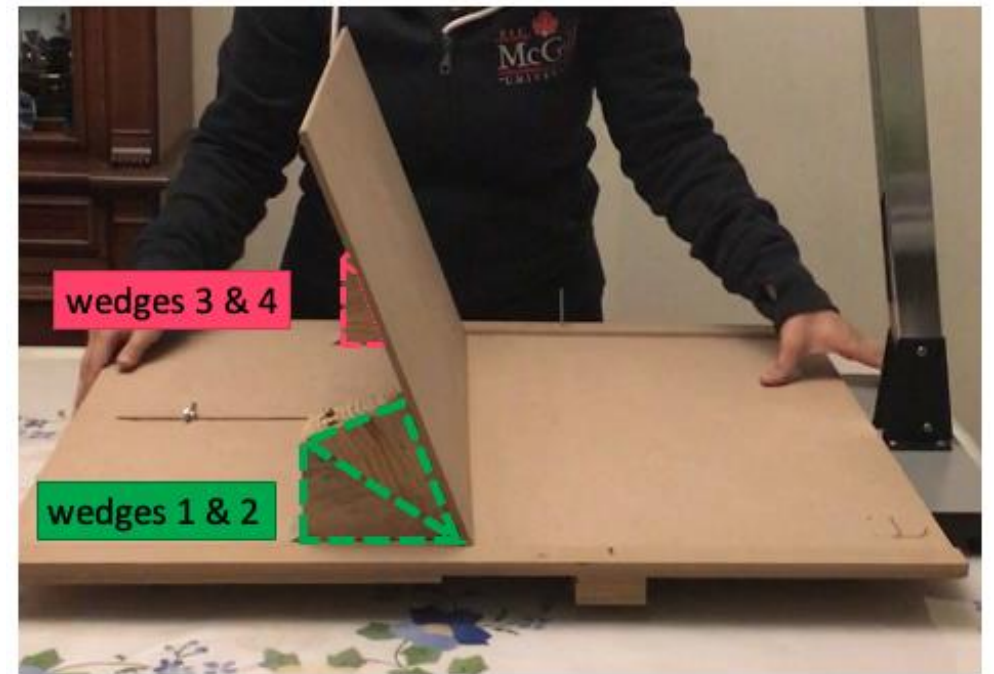
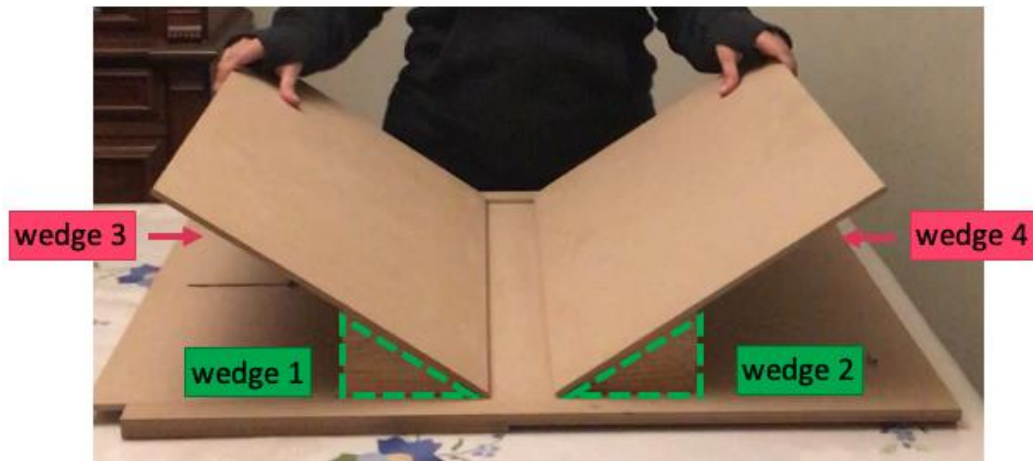
- With **borrowed** (camera & lights) and **built** parts (cradle)
- Advice from various institutions including the *Digital Image Archive of Medieval Music (DIAMM)*



with professional photographer Daniel
Hernández-Salazar

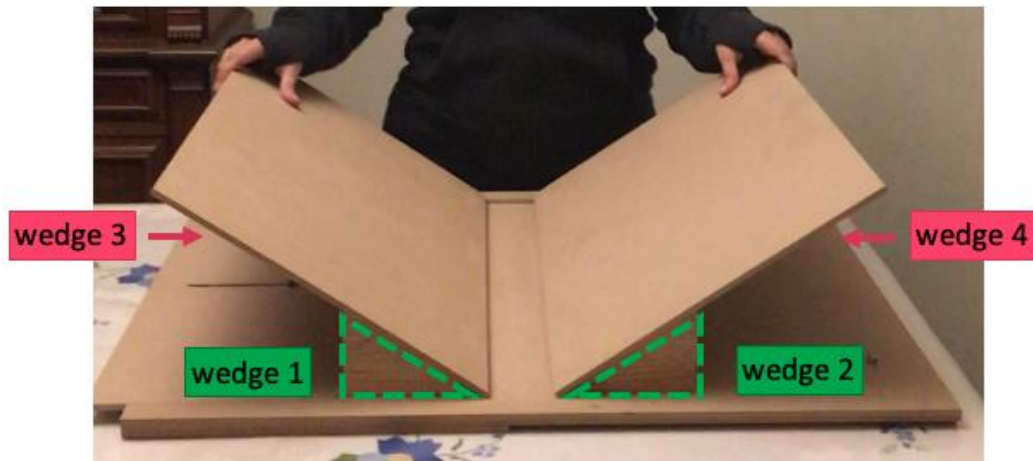
Book Cradle: Configurations

Hold the book opened at 110° angle
by using four wedges of 35°



Book Cradle: Configurations

Hold the book opened at 110° angle
by using four wedges of 35°



German Thomae



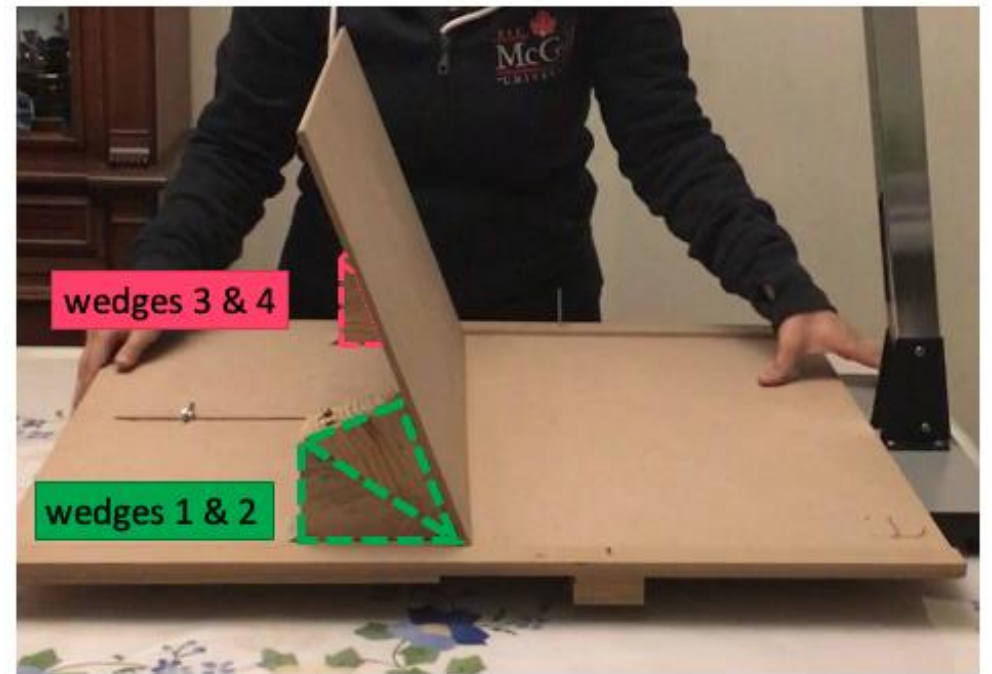
<https://caso.com/services/document-scanning/book-scanning/>

Book Cradle: Configurations

Hold the book opened at 110° angle
by using four wedges of 35°



<https://www.telescopiomania.eu/9161-reproduction-table-bresser-br-cst-copy-stand-40x48-cm-4007922033389.html>



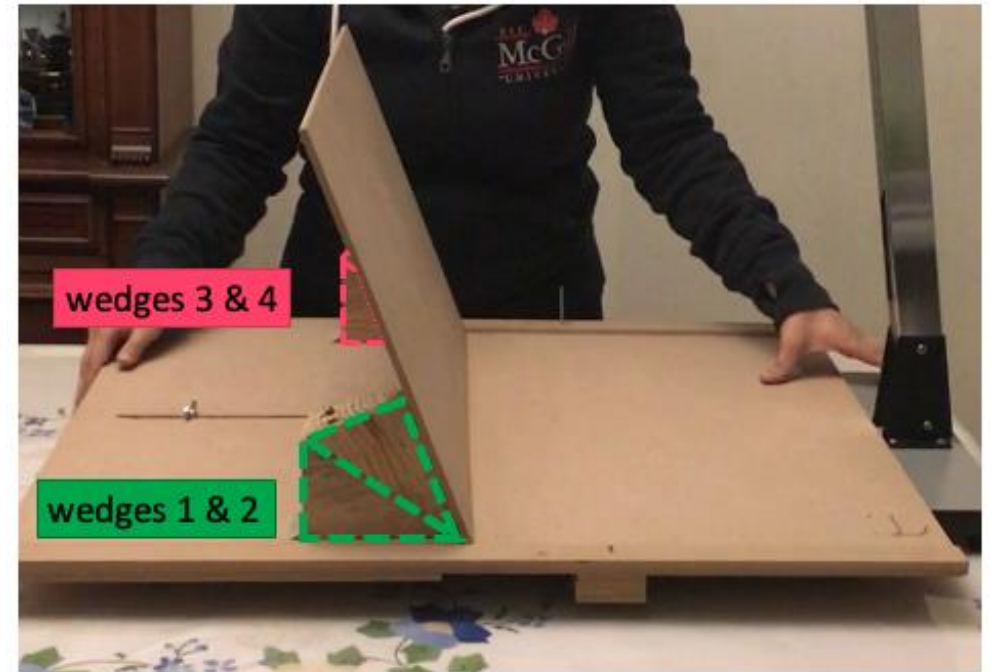
German Thomae

Book Cradle: Configurations

Hold the book opened at 110° angle
by using four wedges of 35°



<https://www.bsb-muenchen.de/einblicke/vom-buch-zum-byte-digitalisierung-an-der-bayerischen-staatsbibliothek>



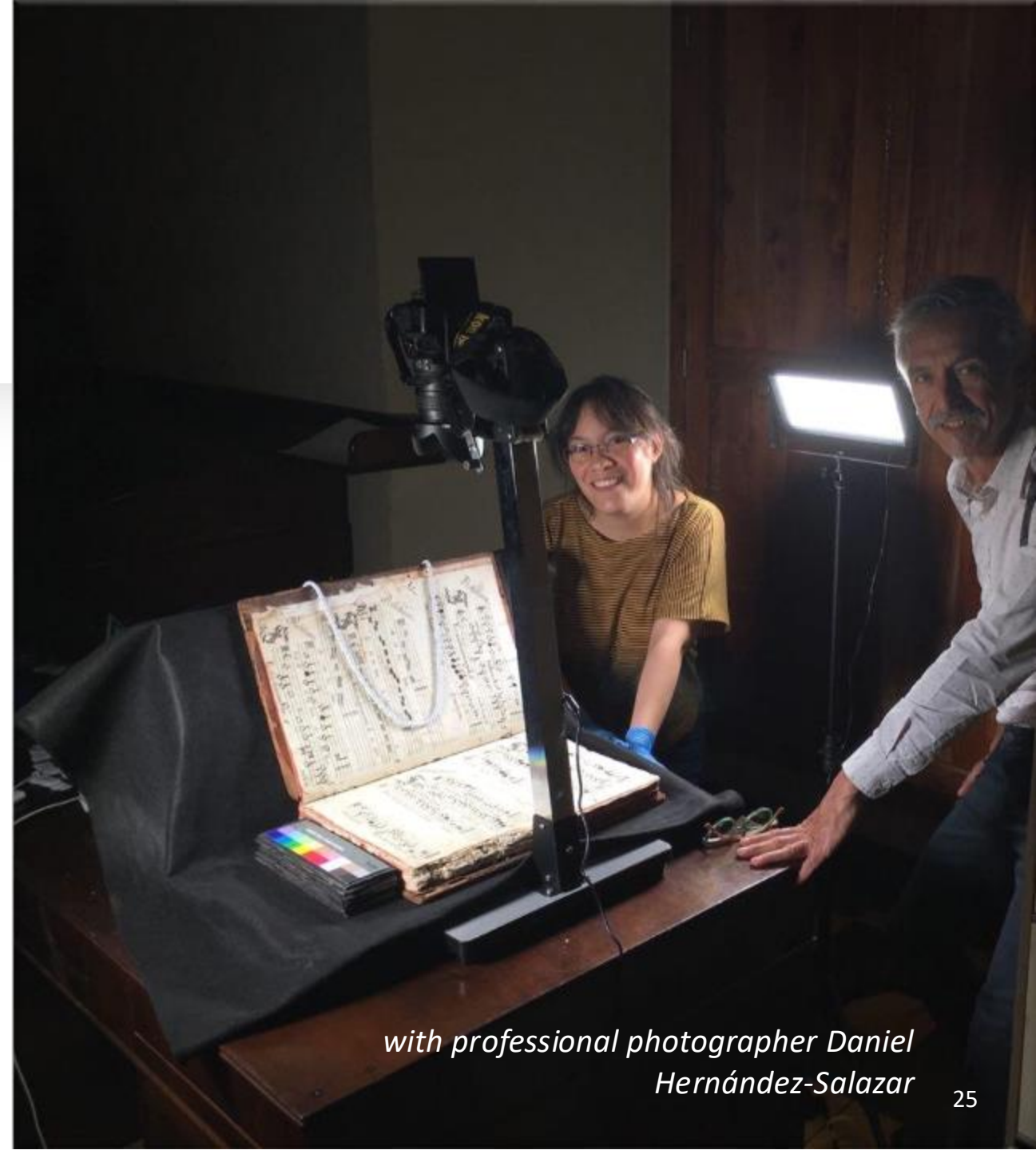
German Thomae



Digitization Stage

AFFORDABLE

- **Equipment:** Do-it-yourself (DIY) book scanner
- **Conservation task:** Outsourced to non-profit institution
- **Digitization task:** Conducted by me with a professional photographer hired for a few of sessions to set up the camera parameters



*with professional photographer Daniel
Hernández-Salazar*

Results



The first staff of music is written in treble clef. It contains a sequence of ten notes: G4 (quarter), F#4 (quarter), E4 (quarter), D4 (quarter), C4 (half), D4 (quarter), E4 (quarter), F#4 (quarter), G4 (quarter), and A4 (half). The staff concludes with a double bar line.

Do mini, in nomine Domini ne,

2. MIR Technologies

Obtaining symbolic files

Optical Music Recognition

What is Optical Music Recognition (OMR)?

- Similar to Optical Character Recognition (OCR)
- OCR recognizes the characters in a digital text document and makes it searchable
- OMR recognizes the music characters in a digital music document and makes the images searchable by their music content

- **OMR is the process of converting images of music documents into symbolic computer representation, such as MIDI, MusicXML, or MEI (Music Encoding Initiative)**



Pixel Domain



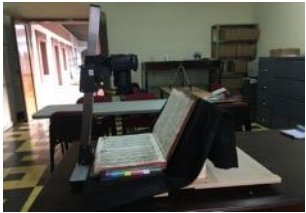
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      <layer xml:id="layer-L6F2N1" num="1">
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        <note xml:id="note-L11F2" dur="2" oct="4" pname="g" accid="n" />
      </layer>
    </staff>
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      <layer xml:id="layer-L6F3N1" num="1">
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          <note xml:id="note-L8F1" dur="16" oct="3" pname="d" accid="s" />
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          <note xml:id="note-L10F1" dur="16" oct="3" pname="f" accid="x" />
        </beam>
        <tuplet>
          <note xml:id="note-L11F1" dur="4" oct="3" pname="a" accid="n" />
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        </tuplet>
      </layer>
    </staff>
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  </measure>
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      <layer xml:id="layer-L13F2N1" num="1">
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      </layer>
    </staff>
    <staff xml:id="staff-L13F3N1" num="2">
      <layer xml:id="layer-L13F3N1" num="1">
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        <note xml:id="note-L15F2" dur="2" oct="2" pname="b" accid="n" />
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Symbolic Domain

Optical Music Recognition (OMR)

MuRET (Music Recognition Encoding and Transcription)

Digitization



Optical music recognition
(OMR)
with human correction



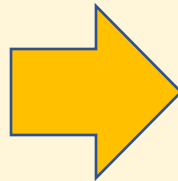
Automatic scoring up
&
editorial correction

MEASURING POLYPHONY

DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC



Digital image



```
<mei>
<music>
<staff n="1">
<clef line="2" shape="G"/>
<note dur="semibrevis" pname="g" oct="5"/>
<note dur="semibrevis" pname="d" oct="5"/>
<dot/>
<note dur="minima" pname="c" oct="5"/>
<ligature form="recta">
<note dur="semibrevis" pname="d" oct="5"/>
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</staff>
</music>
</mei>
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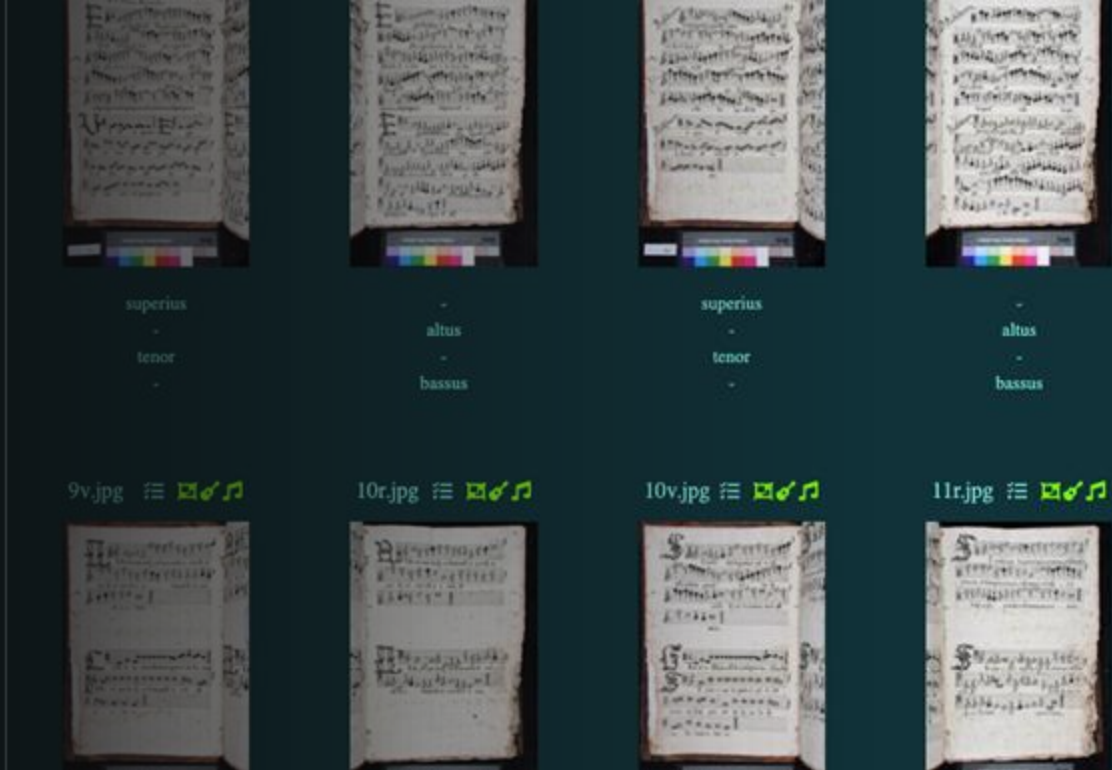
MEI Parts file

encodes the **symbols** in each
part/voice of the images

MuRET

Optical music recognition framework that supports mensural notation

Developed by David Rizo, University of Alicante



MuRET

Music Recognition, encoding, and transcription



Document overview

Logout as martha

Right click on selected images for further exporting and part linking

Unassigned to section

New section...

Move all to default section...



7v.jpg

superius
-
tenor
-

8r.jpg

-
altus
-
bassus

8v.jpg

superius
-
tenor
-

9r.jpg

-
altus
-
bassus

9v.jpg

superius
-
tenor
-

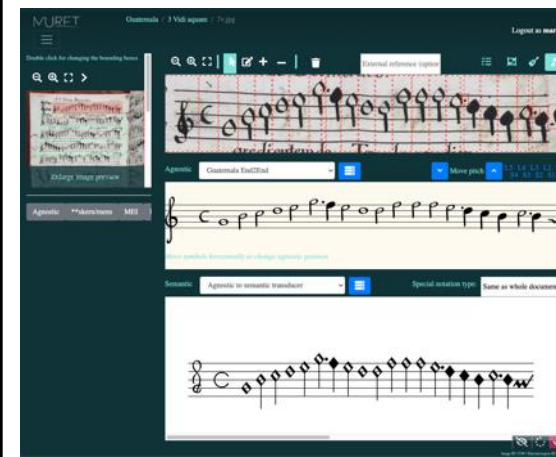
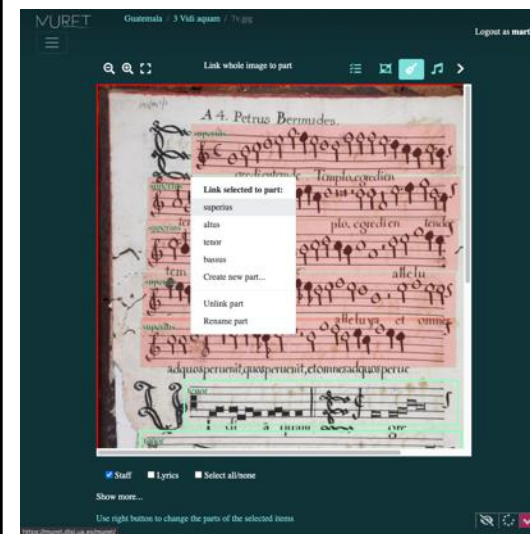
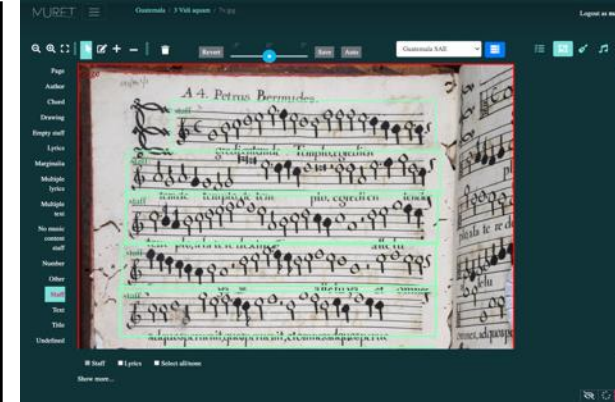
10r.jpg

-
altus
-
bassus

10v.jpg

superius
-
tenor
-

11r.jpg

-
altus
-
bassus



Revert

-5°

0°

5°

Save

Auto

Guatemala SAE



Page
Author
Chord
Drawing
Empty staff
Lyrics
Marginalia
Multiple lyrics
Multiple text
No music content staff
Number
Other
Staff
Text
Title
Undefined

page

893/H51/3

A 4. Petrus Bermudes.

staff

gredientem de Templo, egredien

staff

tem de templo, de tem pio, egredien tencia

staff

tem pio, ala te re dexi

staff

afflu ya et omnes

ad quos peruenit, quos peruenit, et omnes ad quos perue

☒ Staff ☐ Lyrics ☐ Select all/none

MuRET – Interface # 1
OMR STAGE 1: Document Analysis
automatic process



Link whole image to part



895/H5 1/3

A 4. Petrus Bermudes.

superius

superius

superius

tem

superius

superius

adquosperuenit,quosperuenit,etomnesadquosperue

tenor

tenor

allelu

allelu ya et omnes

gredientem de Templo, egredien

plo, egredien tender

Link selected to part:

- superius
- altus
- tenor
- bassus
- Create new part...
- Unlink part
- Rename part

I di a quam ore

☒ Staff

☐ Lyrics

☐ Select all/none

MuRET – Interface # 2

Voice assignation

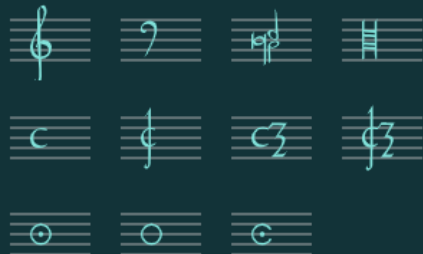
manual process

Double click for changing the bounding boxes

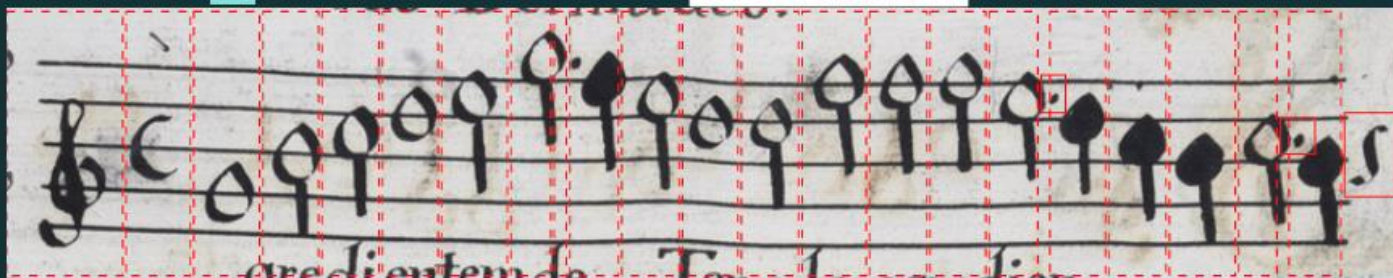


Enlarge image preview

Agnostic **skern/mens MEI PAEC
Clefts | Meters Notes Beamed notes Rests | Accidentals Other



External reference (option)



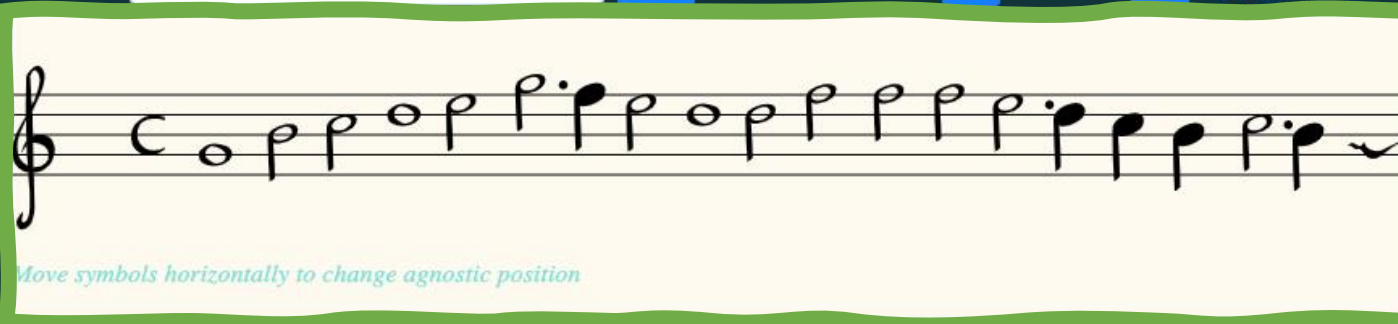
Agnostic

Guatemala End2End



Move pitch

L5 L4 L3 L2 L1
S4 S3 S2 S1



Move symbols horizontally to change agnostic position

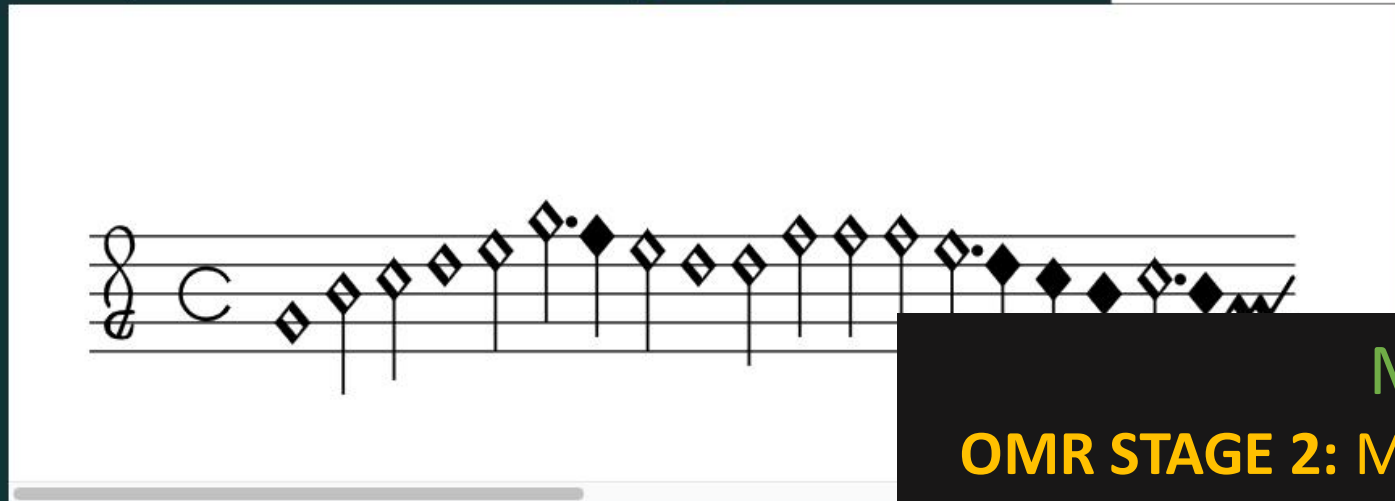
Semantic

Agnostic to semantic transducer



Special notation type:

Same as whole document



agnostic
encoding

MuRET – Interface # 3
OMR STAGE 2: Music Symbol Recognition
automatic process

Double click for changing the bounding boxes



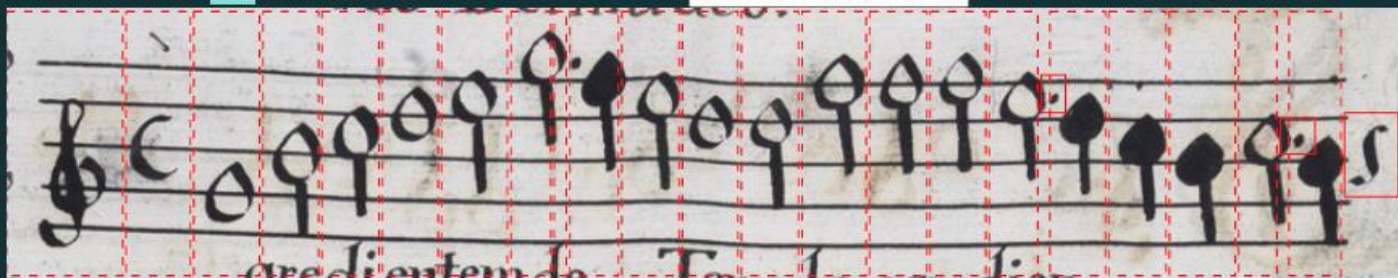
Enlarge image preview

Agnostic ****skern/mens** MEI PAEC

**skm	Ag
*clefG2	2E
*met(C)	2E
Sr_3	2E
sr_3	2E
sg	2E
Ma	2E
Mb\	2E
Mcc	2E
Mdd	2E
see	2E
Mdd	2E
scc	2E
Mhl	2E



External reference (option)



Agnostic

Guatemala End2End



Move pitch



L5 L4 L3 L2 L1
S4 S3 S2 S1



Move symbols horizontally to change agnostic position

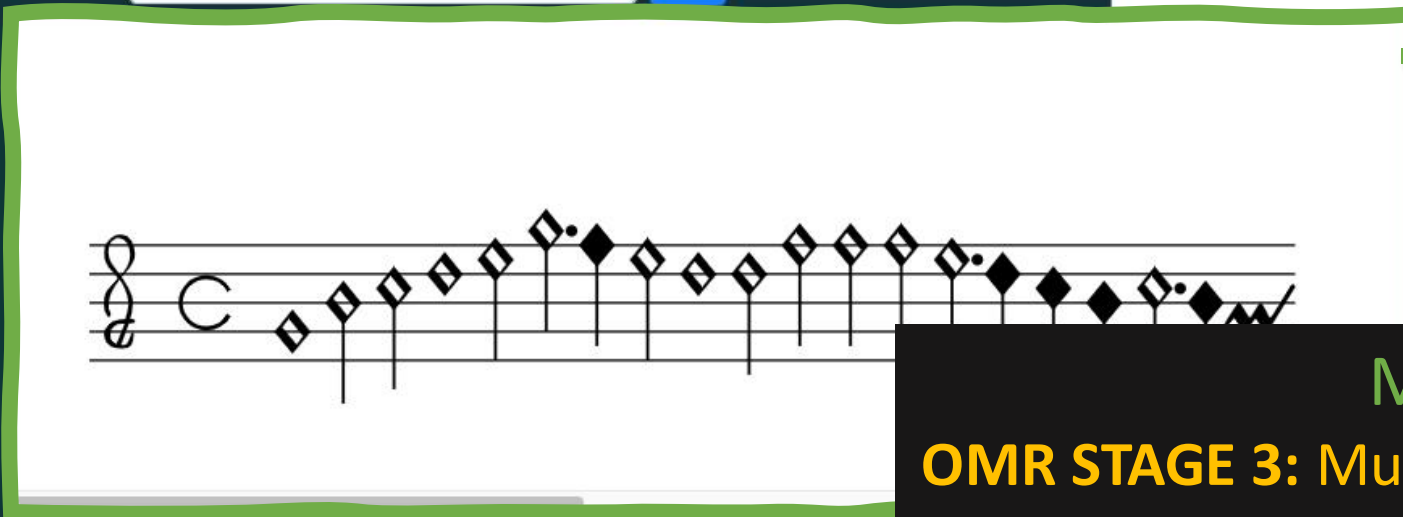
Semantic

Agnostic to semantic transducer



Special notation type:

Same as whole document



semantic
encoding

MuRET – Interface # 3
OMR STAGE 3: Music Notation Recognition
automatic process



Document overview

Logout as martha

Right click on selected images for further exporting and part linking

Unassigned to section

New section...

Move all to default section...



7v.jpg



superius
-
tenor
-

8r.jpg



-
altus
-
bassus

8v.jpg



superius
-
tenor
-

9r.jpg



-
altus
-
bassus

9v.jpg



superius
-
tenor
-

10r.jpg



-
altus
-
bassus

10v.jpg

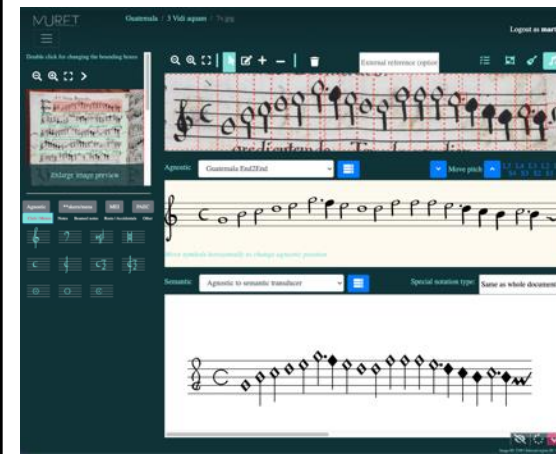
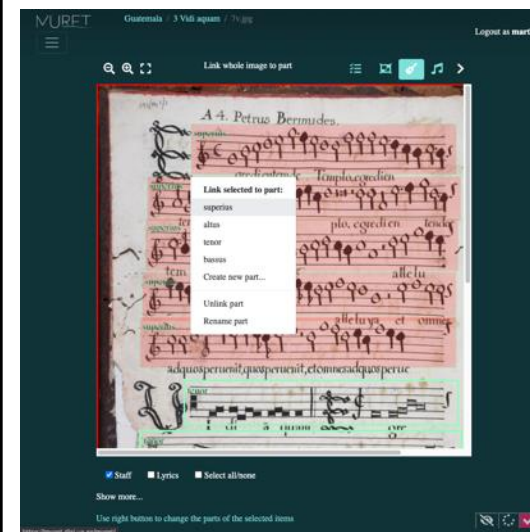
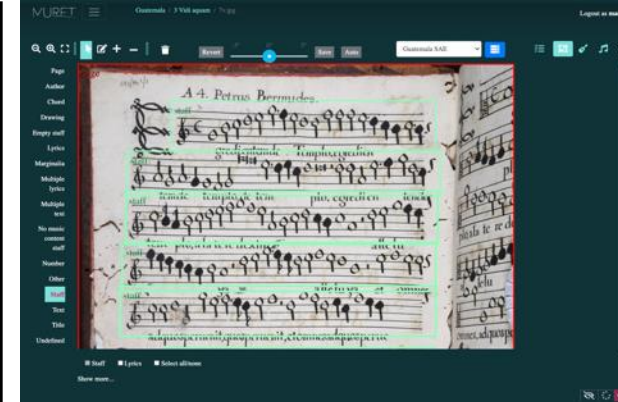


superius
-
tenor
-

11r.jpg



-
altus
-
bassus





Document overview

Right click on selected images for further exporting and part linking

Unassigned to section

New section...

Move all to default section...



7v.jpg

superius
-
tenor
-

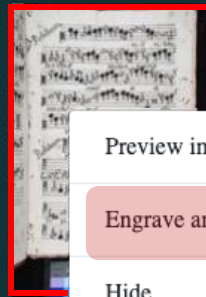
8r.jpg

-
altus
-
bassus

8v.jpg

superius
-
tenor
-

9r.jpg



Preview image...

Engrave and export (MEI)...

Hide

Move to section:

Unassign section

Link whole image to part:

superius

altus

tenor

bassus

Create new part...

Unlink part

9v.jpg

superius
-
tenor
-

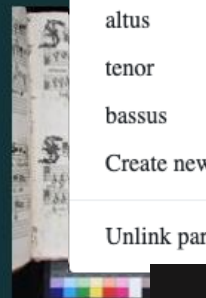
10r.jpg

-
altus
-
bassus

10v.jpg

superius
-
tenor
-

11r.jpg

-
altus
-
bassus

MuRET – Overview Interface

OMR STAGE 4: Music Encoding

automatic process

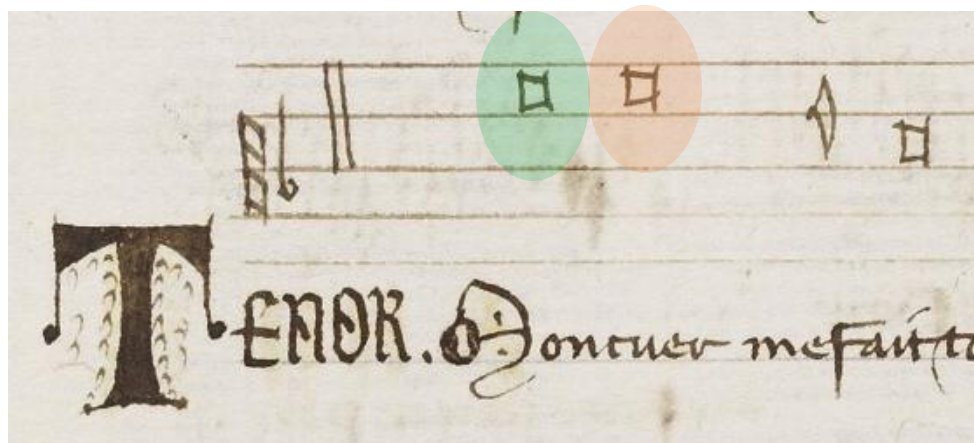
3. MIR Technologies

Obtaining symbolic files

Interpreter of mensural notation
(automatic “scoring up” / “voice alignment”)

What is the Issue with Mensural Notation?

The note shape (the type of note) is not enough to convey the duration of a note.



3 quarter notes

Mon

cuer

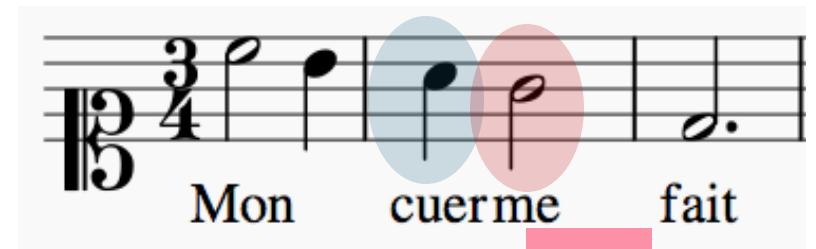
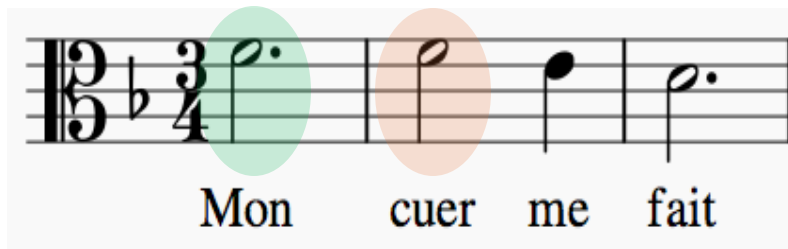
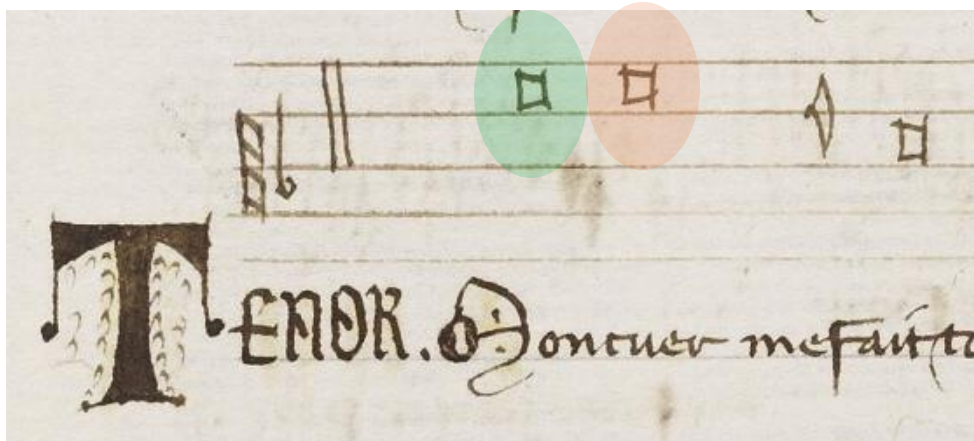
me

fait

2 quarter notes

What is the Issue with Mensural Notation?

The note shape (the type of note) is not enough to convey the duration of a note.



x 2

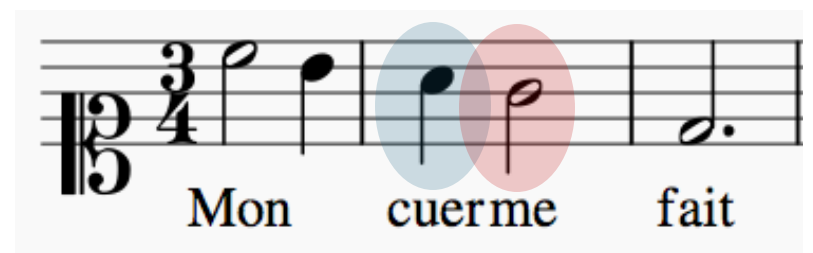
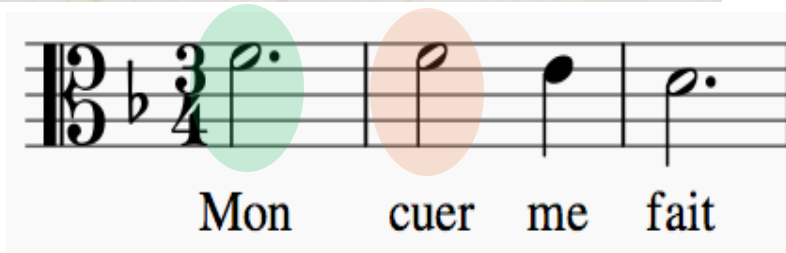
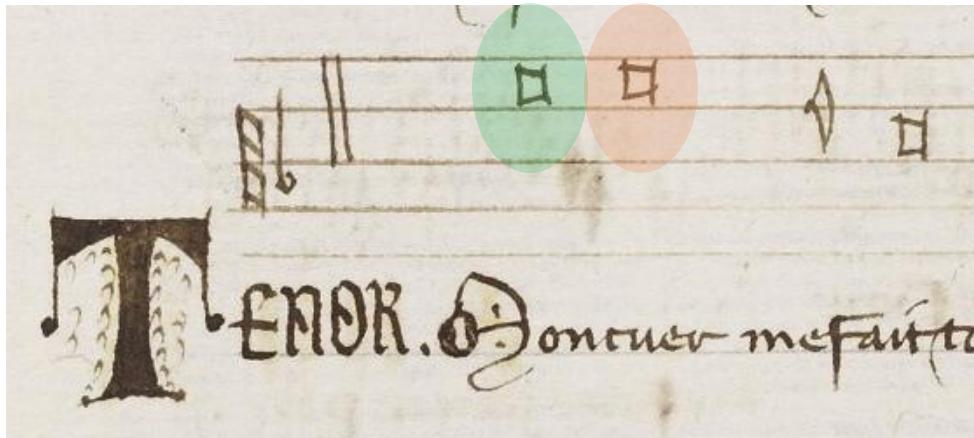
The duration of the individual note symbols in mensural notation depends on

(1) the “mensuration”
or *meter* of the piece,
which can be *duple or triple meter*
(*gives the value of the note by default*)

(2) In triple meter,
the duration of the individual
note symbols is not absolute, but
rather depends on “context”

What is the Issue with Mensural Notation?

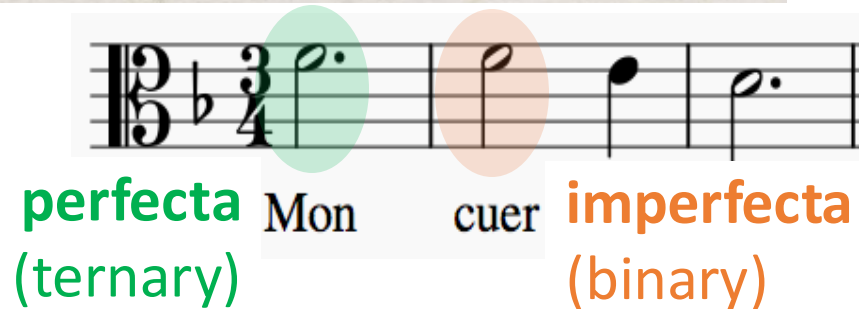
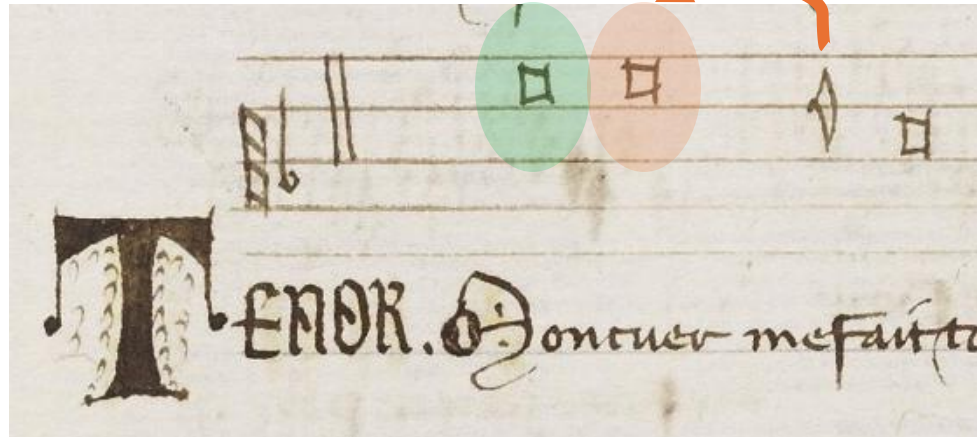
In **triple meter**, the note shape is not enough to convey the duration of a note. **The note's default duration can be modified by the context** (i.e., notes preceding or following)



What is the Issue with Mensural Notation?

In **triple meter**, the note shape is not enough to convey the duration of a note. **The note's default duration can be modified by the context** (i.e., notes preceding or following)

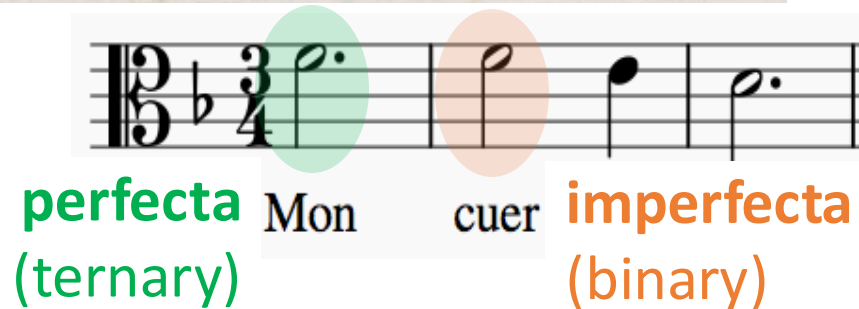
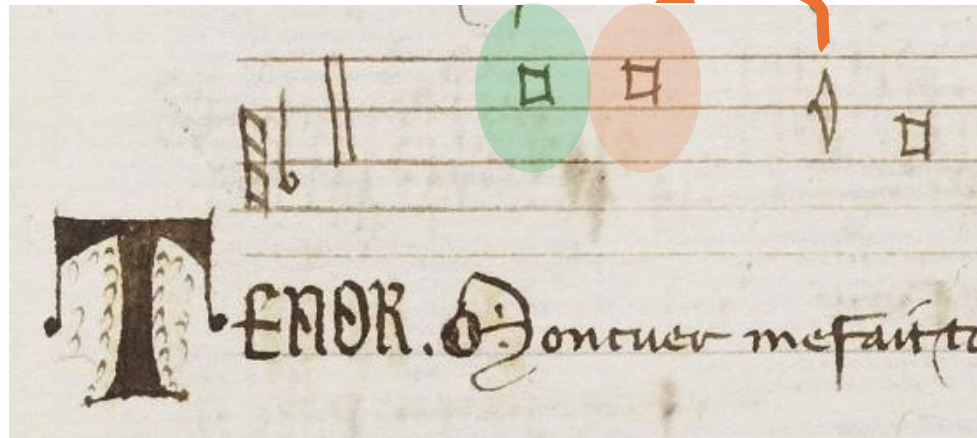
IMPERFECTION



What is the Issue with Mensural Notation?

In **triple meter**, the note shape is not enough to convey the duration of a note. **The note's default duration can be modified by the context** (i.e., notes preceding or following)

IMPERFECTION



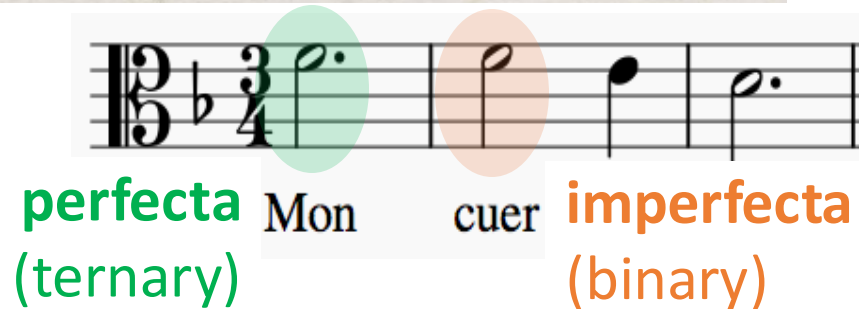
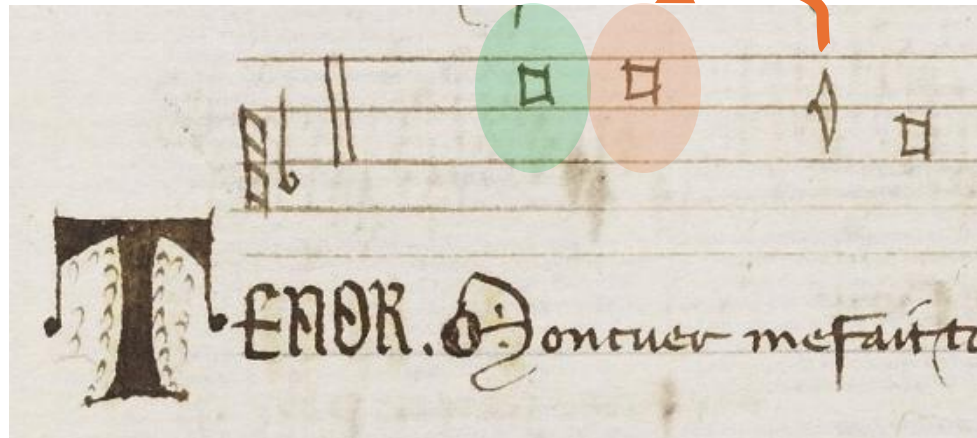
ALTERATION



What is the Issue with Mensural Notation?

Principles of Imperfection and Alteration

IMPERFECTION



ALTERATION



3. MIR Technologies

Obtaining symbolic files

Interpreter of mensural notation
(automatic “scoring up” / “voice alignment”)

- Developed an expert system based on the “principles of imperfection and alteration”
- Implemented it into Python (Master’s degree)
- Re-implement it as a part of an online mensural notation editor, which is **the next tool used**

Automatic Voice Alignment & Editorial Correction

Measuring Polyphony Editor (MP-Editor)

Digitization



Optical music recognition
(OMR)
with human correction



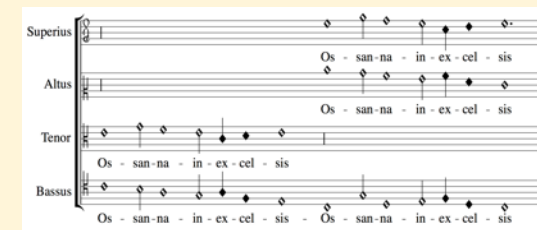
Automatic scoring up
&
editorial correction

MEASURING POLYPHONY

DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC

```
<mei>
<music>
<staff n="1">
  <clef line="2" shape="G"/>
  <note dur="semibrevis" pname="g" oct="5"/>
  <note dur="semibrevis" pname="d" oct="5"/>
  <dot/>
  <note dur="minima" pname="c" oct="5"/>
  <ligature form="recta">
    <note dur="semibrevis" pname="d" oct="5"/>
    <note dur="semibrevis" pname="g" oct="4"/>
  </ligature>
  <note dur="brevis" pname="b" oct="4"/>
</staff>
</music>
</mei>
```

MEI Parts file
encodes the **symbols** in each
part/voice of the images



MEI Score file
interpreted durations
(i.e., *alteration & imperfection*)

Measuring Polyphony Editor

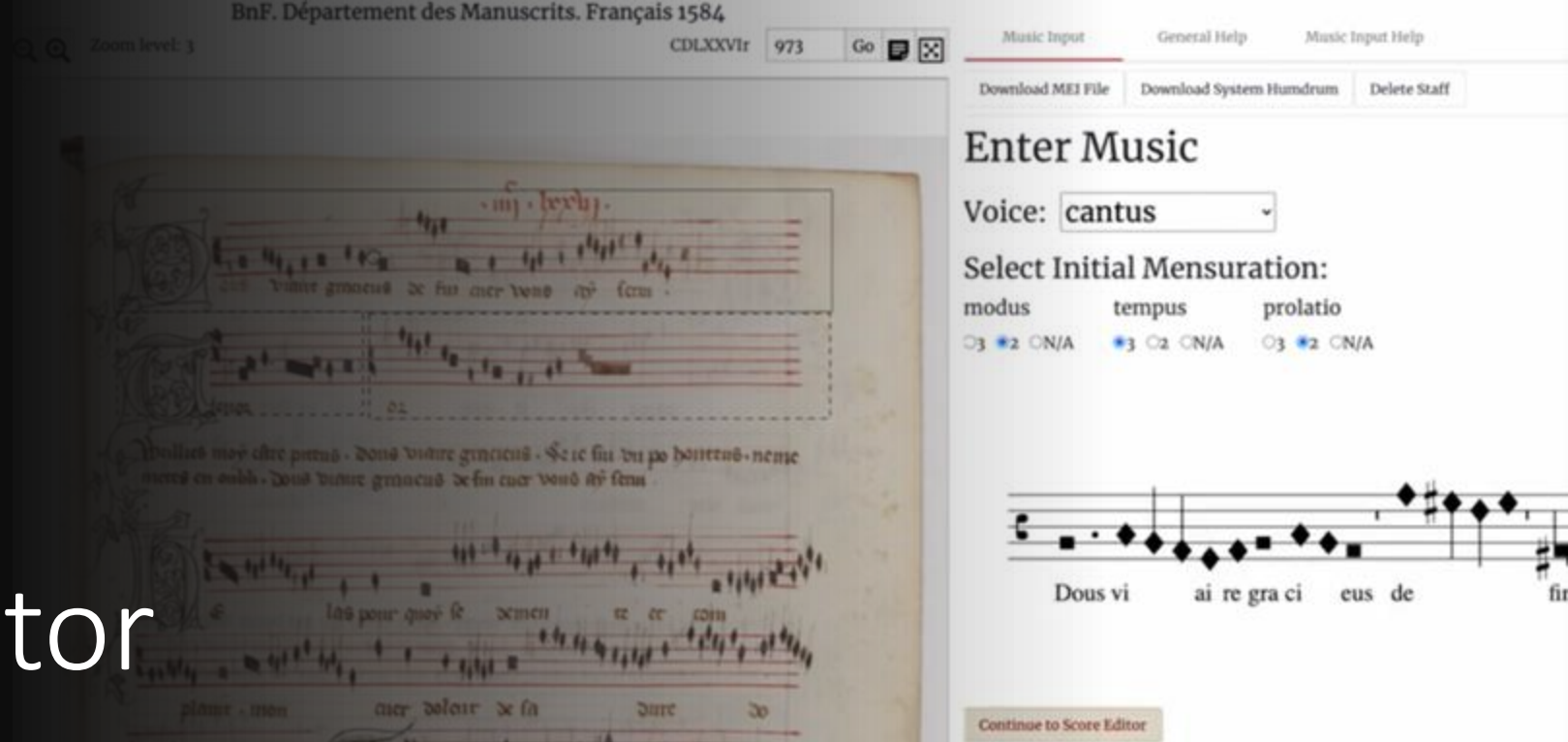
Online editor for mensural notation

Karen Desmond (PI), Maynooth University

Mostly developed at McGill University:

- **Juliette Regimbal** (lead developer)
- **Martha E. Thomae** (automatic voice alignment function)

And with **Laurent Pugin** (Verovio), **Craig Sapp** (Humdrum), and **Andrew Hankinson** (IILF)



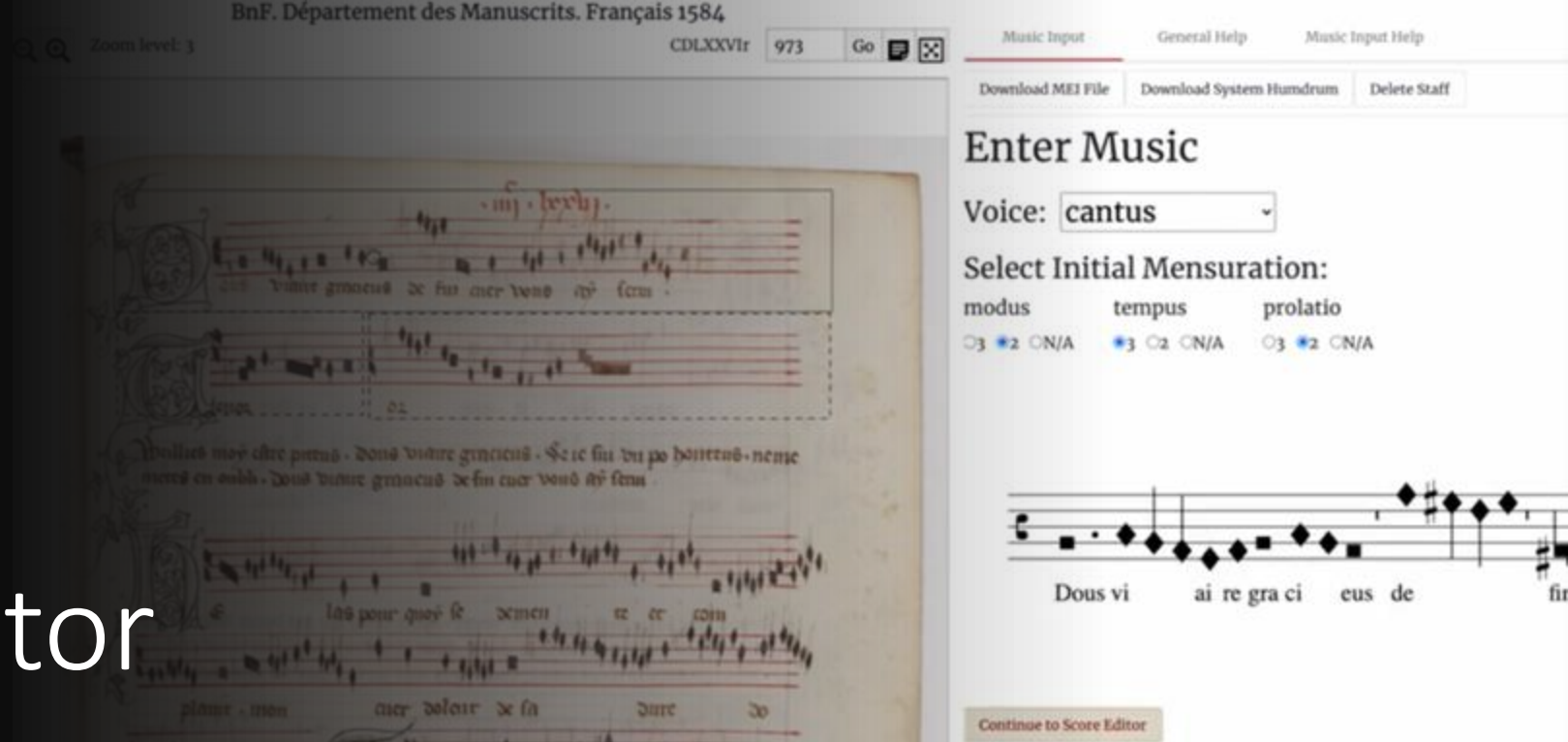
MEASURING POLYPHONY

DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC

Measuring Polyphony Editor

Online editor for mensural notation with **two parts**:

1. **Input Editor**: The user types in the notes using the computer keyboard
2. **Score Editor**: It processes the notes entered, computes their duration based on a system of rules, and lines up the piece in score layout



MEASURING POLYPHONY

MuRET
Editor
editor.measuringpolyphony.org/#/
CAQ
McGill's Services...
Software to Downl...
Verovio Editor
Manuscriptorium
MEI Garage
Verovio MEI Viewer
MuRET
Editor
MEC 2019 VHV W...
MEC 2019 Tutorial...
Welcome to PEM [...
Welcome to Spani...
Other Bookmarks


MEASURING POLYPHONY
DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC

BROWSE
TRANSCRIBE
ABOUT

Load a manuscript

Manuscript Repository: Gallica URL:

Submit

For example, to transcribe a piece from the famous *Roman de Fauvel* manuscript, choose *Gallica* in the dropdown and paste this URL:

<https://gallica.bnf.fr/ark:/12148/btv1b8454675g>

Load an .mei file

Compatible Parts-MEI File: Choose File No file chosen

Load File

The Measuring Polyphony Music Editor (the MP Editor), funded by the [National Endowment for the Humanities](#), is a prototype web application that allows users with no expertise in music encoding to transcribe and digitally encode transcriptions of polyphony in mensural notation, and create links between these music transcriptions to zones of the digital images of the original medieval manuscripts. To download a White Paper on the MP Editor, click [here](#). For more information about the Measuring Polyphony project, click [here](#).

Project Team

Karen Desmond, Brandeis University, Massachusetts, USA: Principal Investigator & Project Director
Juliette Regimbal, McGill University, Canada: Prototype developer
Martha Thomae, McGill University, Canada: Prototype developer
Andrew Hankinson, RISM, Switzerland: Consultant on diva.js, MEI
Laurent Pugin, University of Bern & RISM, Switzerland: Consultant on Verovio, MEI
Craig Sapp, Stanford University, California, USA: Consultant on Humdrum, MEI

Student Research Assistants

Caelan Gadweh-Meaden, Brandeis University, USA
Zen Kuriyama, Brandeis University, USA
Daniel Shapiro, Brandeis University, USA

With thanks also to Julie Cumming and her music palaeography students at McGill University - Christina Colanduoni, Mayu Egan, Geneviève Gates-Panneton, Namgon Lee, and Celia Morin - for testing the MP Editor in December 2020.

Advisory Board

Julie Cumming, McGill University, Canada
Julia Craie McEelvy, University of Oxford, UK

Load a manuscript

Manuscript Repository: URL:

Submit

For example, to transcribe a piece from the famous *Roman de Fauvel* manuscript, choose *Gallica* in the dropdown and paste this URL:

<https://gallica.bnf.fr/ark:/12148/btv1b8454675g>

Load an .mei file

Compatible Parts-MEI File: No file chosen

Load File

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Project Team

Karen Desmond, Brandeis University, Massachusetts, USA: Principal Investigator & Project Director

INPUT EDITOR

The notes were not typed in as the loaded file already had the symbols recognized in the previous step

editor.measuringpolyphony.org/#input/iiif/https%25253A%25252F%25252Fmuret.dlsi.ua.es%25252Fiiif%25252F6%25252F223%25252Fmanifest.json

MEASURING POLYPHONY
DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC

Pedro Bermudez

Zoom level: 3

166r.jpg - 167r.jpg 166v Go

Music Input General Help Music Input

Download MEI File Download System Humdrum Del

Enter Music

Voice:

Select Initial Mensuration:

modus	tempus	prolatio
☐ 3 ☒ 2 ☐ N/A	☐ 3 ☒ 2 ☐ N/A	☒ 3 ☐ 2 ☐ N/A

Mensuration Sign:

166v.jpg p1 s1

Continue to Score Editor

modus

☐ 3 ☒ 2 ☐ N/A

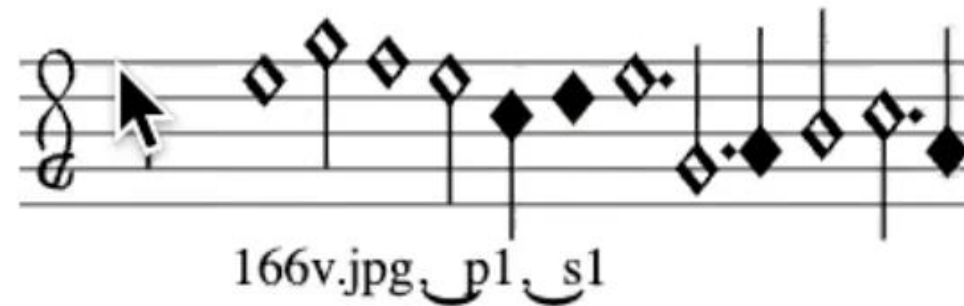
tempus

☐ 3 ☒ 2 ☐ N/A

prolatio

☒ 3 ☐ 2 ☐ N/A

Mensuration Sign: C3 ▾



[Continue to Score Editor](#)

SCORE EDITOR

Automatic lining up of the voices into a score

Can provide editorial corrections
And export into an MEI Score file

[illegible]

Score Editor

Score Editor Help

Return to Music Input

Download MEI File(s)

Copy MEI Score to Clipboard

16 Missa de Bomba

superius

altus

tenor

bassus

The image displays a musical score for a four-part setting of a mass, titled "16 Missa de Bomba". The score is written for four voices: superius, altus, tenor, and bassus. Each staff begins with a treble clef (for superius and altus) or a bass clef (for tenor and bassus), followed by a common time signature (C) and a 3/4 time signature (3). The notation includes various musical symbols such as notes, rests, and accidentals. Below the staves, there are labels for image placeholders: "166v.jpg_p1_s1", "166v.jpg_p1_s2", "167r.jpg_p1_s1", "167r.jpg_p1_s2", "166v.jpg_p1_s4", "166v.jpg_p1_s5", "167r.jpg_p1_s4", and "167r.jpg_p1_s5".

Bar by: semibreve ☐ Switch to Modern Clefs ☐ Add Dissonance Labels ☐

Continue in Editorial Mode

SCORE EDITOR

Automatic lining up of the voices into a score

Can provide editorial corrections
And export into an MEI Score file

MuRET

Editor

[editor.measuringpolyphony.org/#/score/iiif/https%2523A%25252F%25252Fmuret.dlsi.ua.es%25252Fiiif%25252F6%25252F](#)

CAQ
 McGill's Services...
 Software to Down...
 Verovio Editor
 Manuscriptorium
 MEI Garage
 Verovio MEI Viewer

MEASURING POLYPHONY
DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC

Pedro Bermudez

Zoom level: 3
166r.jpg - 167r.jpg
Go

Score Editor
Score Editor Help

Return to Music Input
Download MEI File(s)

superius
altus
tenor
bassus

166v
KODAK Color Control Patches

Bar by: semibreve Switch to Modern Clefs ☐

Continue in Editorial Mode

16 Missa de Bomba



superius

altus

tenor

bassus

166v.jpg_p1_s1

166v.jpg_p1_s2

167r.jpg_p1_s1

167r.jpg_p1_s2

166v.jpg_p1_s4

166v.jpg_p1_s5

167r.jpg_p1_s4

167r.jpg_p1_s5

Bar by: Switch to Modern Clefs ☐ Add Dissonance Labels ☐

[Continue in Editorial Mode](#)

SCORE EDITOR

Automatic lining up of the voices into a score

Can provide editorial corrections
And export into an MEI Score file

MuRET Editor

editor.measuringpolyphony.org/#/score/iiif/https%25253A%25252F%25252Fmuret.dlsi.ua.es%25252Fiiif%25252F6%25252F

CAQ McGill's Services... Software to Downl... Verovio Editor Manuscriptorium MEI Garage Verovio MEI Viewer

MEASURING POLYPHONY
DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC

Pedro Bermudez

Zoom level: 3 166r.jpg - 167r.jpg Go

Score Editor Score Editor Help

Return to Music Input Download MEI File(s) Copy MEI Score to Clipboard



166v

KODAK Color Control Patches

16 Missa de Bomba

superius

altus

tenor

bassus

166v.jpg_p1_s1 166v.jpg_p1_s2

167r.jpg_p1_s1 167r.jpg_p1_s2

166v.jpg_p1_s4 166v.jpg_p1_s5

167r.jpg_p1_s4 167r.jpg_p1_s5

Bar by: semibreve Switch to Modern Clefs Add Dissonance Labels

Continue in Editorial Mode

SCORE EDITOR

Automatic lining up of the voices into a score

Can provide editorial corrections
And export into an MEI Score file

MuRET Editor

editor.measuringpolyphony.org/#/score/iiif/https%25253A%25252F%25252Fmuret.dlsi.ua.es%25252Fiiif%25252F6%25252F

CAQ McGill's Services... Software to Downl... Verovio Editor Manuscriptorium MEI Garage Verovio MEI Viewer

MEASURING POLYPHONY
DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC

Pedro Bermudez

Zoom level: 3 166r.jpg - 167r.jpg Go

Score Editor Score Editor Help

Return to Music Input Download MEI File(s) Copy MEI Score to Clipboard



166v

KODAK Color Control Patches

16 Missa de Bomba

superius

altus

tenor

bassus

166v.jpg_p1_s1 166v.jpg_p1_s2

167r.jpg_p1_s1 167r.jpg_p1_s2

166v.jpg_p1_s4 166v.jpg_p1_s5

167r.jpg_p1_s4 167r.jpg_p1_s5

Bar by: semibreve Switch to Modern Clefs Add Dissonance Labels

Continue in Editorial Mode

SCORE EDITOR

Automatic lining up of the voices into a score

Can provide editorial corrections
And export into an MEI Score file

MuRET Editor

editor.measuringpolyphony.org/#/score/iiif/https%25253A%25252F%25252Fmuret.dlsi.ua.es%25252Fiiif%25252F6%25252F...
CAQ McGill's Services... Software to Downl... Verovio Editor Manuscriptorium MEI Garage Verovio MEI Viewer

MEASURING POLYPHONY
DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC

Pedro Bermudez

Zoom level: 3 166r.jpg - 167r.jpg Go

Score Editor Score Editor Help

Return to Music Input Download MEI File(s) Copy MEI Score to Clipboard



166v

KODAK Color Control Patches

16 Missa de Bomba

superius

166v.jpg_p1_s1 166v.jpg_p1_s2

altus

167r.jpg_p1_s1 167r.jpg_p1_s2

tenor

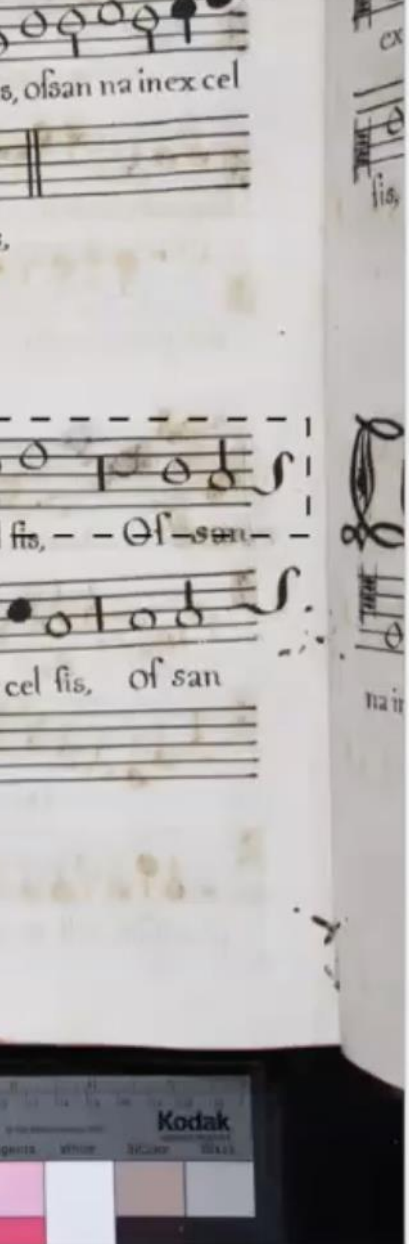
166v.jpg_p1_s4 166v.jpg_p1_s5

bassus

167r.jpg_p1_s4 167r.jpg_p1_s5

Bar by: semibreve Switch to Modern Clefs Add Dissonance Labels

Continue in Editorial Mode



superius

altus

tenor

bassus

SCORE EDITOR

Automatic lining up of the voices into a score

Can provide editorial corrections

And export into an MEI Score file

166v.jpg_p1_s4

166v.jpg_p1_s4

167r.jpg_p1_s4

167r.jpg_p1_s5

Bar by: **semibreve** ☒ Switch to Modern Clefs ☐ Add Dissonance Labels ☐

Continue in Editorial Mode

SCORE EDITOR

Automatic lining up of the voices into a score
Can provide editorial corrections
And export into an MEI Score file

BROWSE TRAN

167r.jpg

Go



Score Editor

Score Editor Help

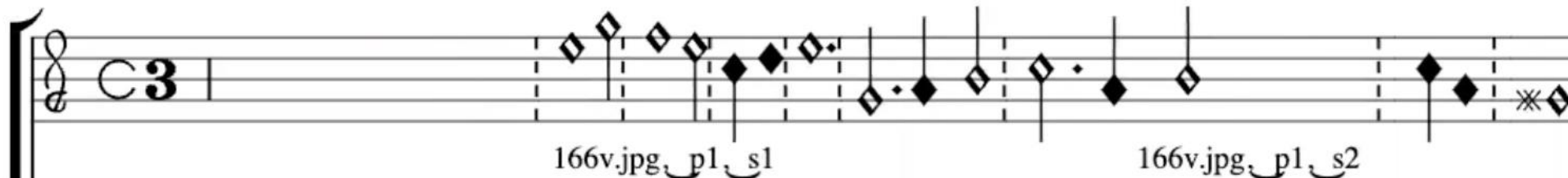
Return to Music Input

Download MEI File(s)

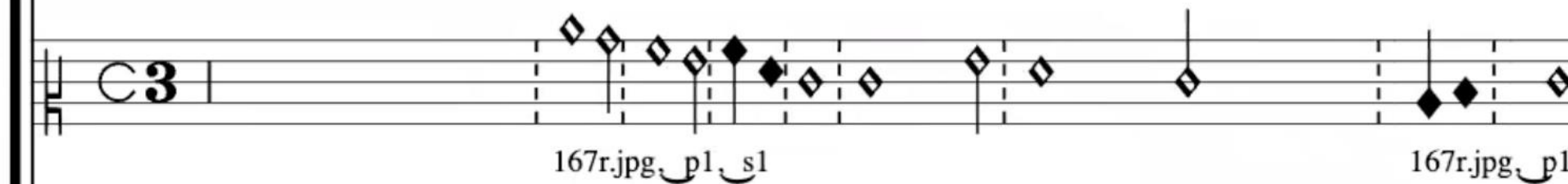
Copy MEI Score to Clipboard

16 Missa de Bomba

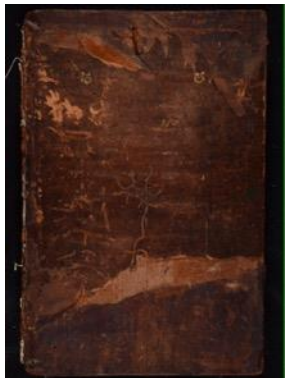
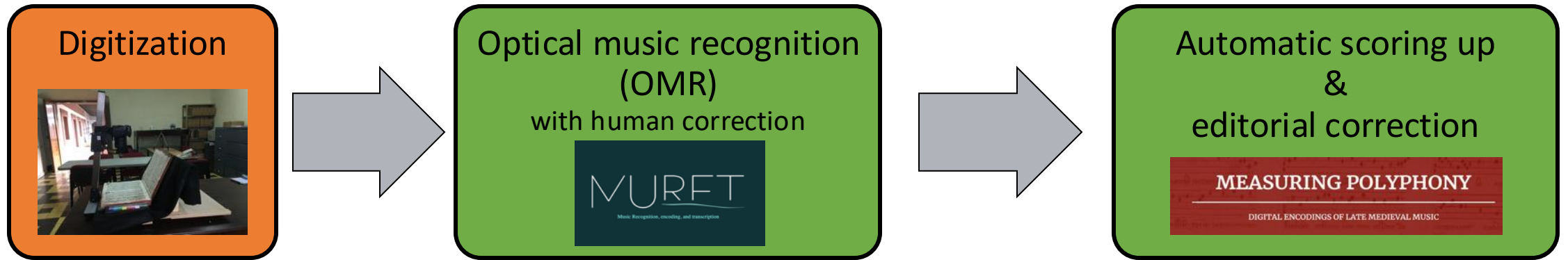
superius



altus



Digitization & MIR Pipeline



Manuscript



Digital images

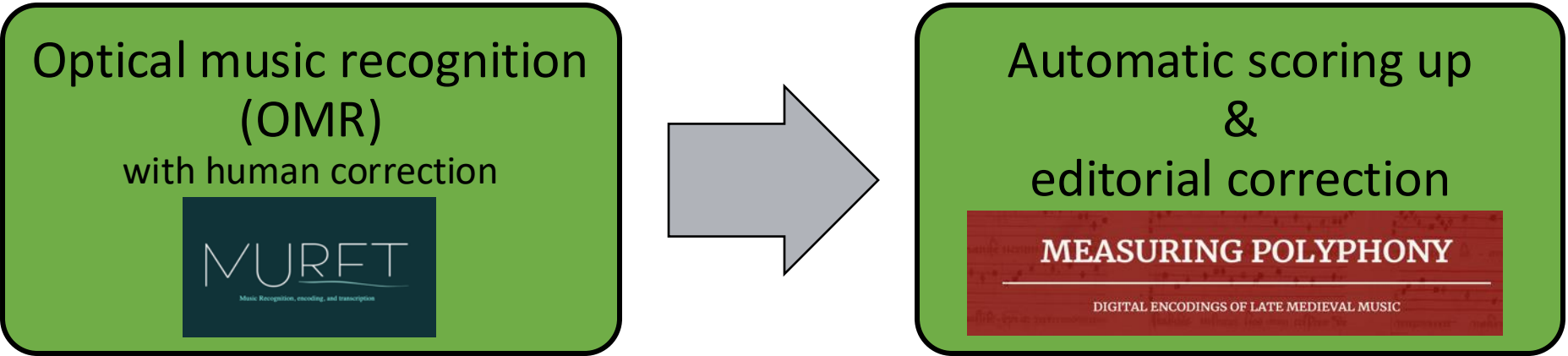
```
<mei>
<music>
<staff n="1">
  <clef line="2" shape="G"/>
  <note dur="semibrevis" pname="g" oct="5"/>
  <note dur="semibrevis" pname="d" oct="5"/>
  <dot/>
  <note dur="minima" pname="c" oct="5"/>
  <ligature form="recta">
    <note dur="semibrevis" pname="d" oct="5"/>
    <note dur="semibrevis" pname="g" oct="4"/>
  </ligature>
  <note dur="brevis" pname="b" oct="4"/>
</staff>
</music>
</mei>
```

MEI Parts file
encodes the **symbols** in each
part/voice of the images



MEI Score file
interpreted durations
(i.e., *alteration & imperfection*)

Get the Best of Both Tools!



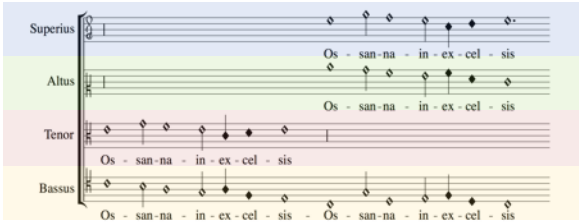
```
<mei>
<music>
  <staff n="1">
    <clef line="2" shape="G"/>
    <note dur="semibrevis" pname="g" oct="5"/>
    <note dur="semibrevis" pname="d" oct="5"/>
    <dot/>
    <note dur="minima" pname="c" oct="5"/>
    <ligature form="recta">
      <note dur="semibrevis" pname="d" oct="5"/>
      <note dur="semibrevis" pname="g" oct="4"/>
    </ligature>
    <note dur="brevis" pname="b" oct="4"/>
  </staff>
</music>
</mei>
```

MEI Parts file
encodes the **symbols**
in each **part/voice**

Interoperability between MuRET & MP Editor		
David Rizo, Juliette Regimbal, and Martha Thomae		
	MuRET	MP Editor
Note shapes (and other symbols)	Automatic recognition	Manual entry (in input editor by computer keyboard)

Note duration
Manual entry
[\(in semantic encoding, stage 3\)](#)

Automatic computation



MEI Score file
interpreted durations
(i.e., *alteration & imperfection*)

Double click for changing the bounding boxes



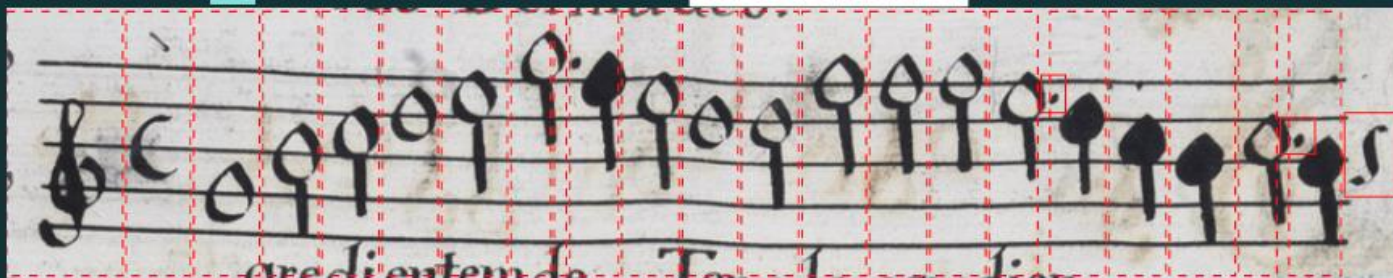
Enlarge image preview

Agnostic ****skern/mens** MEI PAEC

**skm	Ag	+
*clefG2	2E	+
*met(C)	2E	-
Sr_3	2E	
sr_3	2E	
sg	2E	
Ma	2E	
Mb\	2E	
Mcc	2E	
Mdd	2E	
see	2E	
Mdd	2E	
scc	2E	
Mhl	2E	



External reference (option)



Agnostic

Guatemala End2End



Move pitch



L5 L4 L3 L2 L1
S4 S3 S2 S1



Move symbols horizontally to change agnostic position

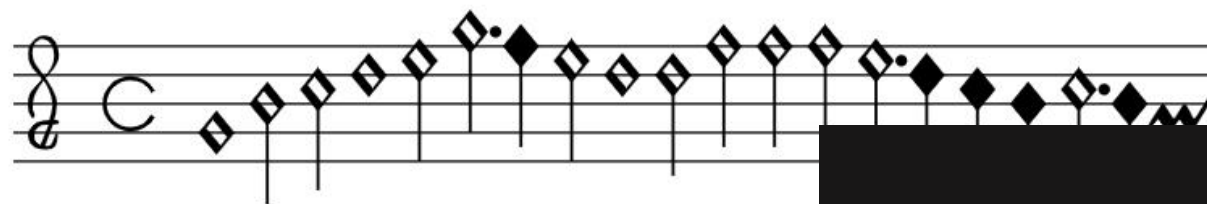
Semantic

Agnostic to semantic transducer



Special notation type:

Same as whole document



Inserting the note
durational quality
(perfecta, imperfecta,
or altera) requires
manual endoding

MuRET – Interface # 3

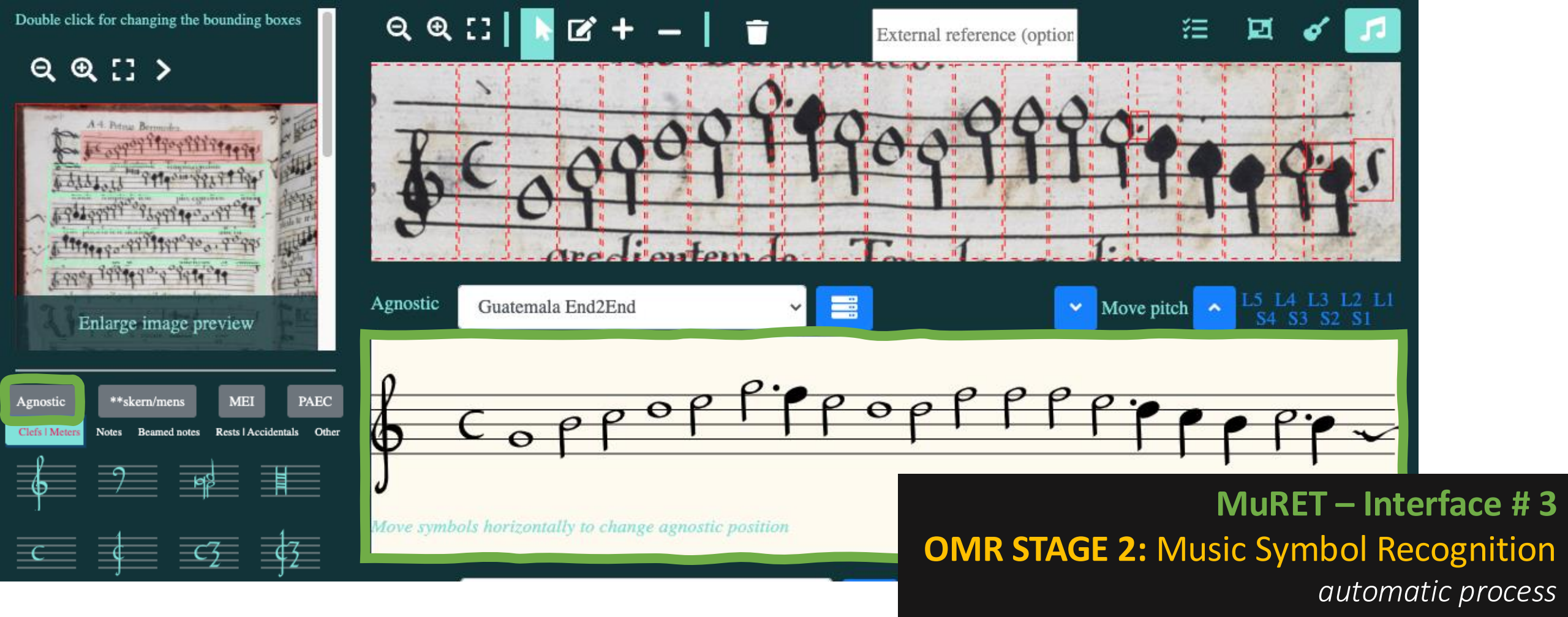
OMR STAGE 3: Music Notation Recognition

automatic process

Human Correction After Each Automatic Processes

“Symbol recognition” in *MuRET*
& “Scoring up” in *MP Editor*

How to facilitate this?



MuRET already has an interface that facilitates spotting and correcting errors in the “symbol recognition” process

The recognized symbols are lined up with the image
& The user can add/delete/modify the symbols

What about the MP Editor?

How can we facilitate human correction here?

The corrections done here are **editorial** in nature, we correct **errors made by the scribe**

How can we make it easier to spot these errors?

SCORE EDITOR

Automatic lining up of the voices into a score

Can provide editorial corrections

And export into an MEI Score file

MEASURING POLYPHONY
DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC

Pedro Bermudez

Zoom level: 3 152v.jpg - 153r.jpg Go

Score Editor Score Editor Help

Return to Music Input Download MEI File(s) Copy MEI Score to Clipboard

16 Missa de Bomba

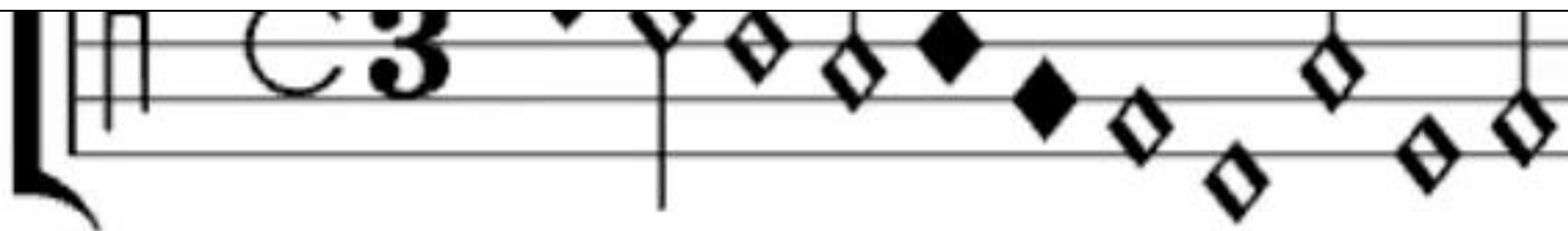
superius
altus
tenor
bassus

166v.jpg_p1_s1 166v.jpg_p1_s2 166v.jpg_p1_s3
167r.jpg_p1_s1 167r.jpg_p1_s2 167r.jpg_p1_s3
166v.jpg_p1_s4 166v.jpg_p1_s5 166v.jpg
167r.jpg_p1_s4 167r.jpg_p1_s5

Bar by: none Switch to Modern Clef Add Dissonance Labels

Continue in Editorial Mode

bassus



167r.jpg, p1, s4

Bar by: none 

Switch to Modern Clef ☐

Add Dissonance Labels ☐

Continue in Editorial Mode

“Add Dissonance Labels” Filter

Labels the different types of dissonances according to their functions

Dissonances that function as passing tones (P/p), neighbour tones (N/n), suspensions (s), etc.

& dissonances whose functionality is not known

→ possible scribal errors?

superius

altus

tenor

bassus

Bar by: none ▾

Switch to Modern Clef ☐

Add Dissonance Labels ☐

Continue in Editorial Mode

Interoperability between MP Editor & Dissonance Filter (Humlib)

Craig Sapp, Juliette Regimbal, and Martha Thomae

The image displays a musical score for four voices: superius, altus, tenor, and bassus. The notation is in C-clef and common time (C). The score is annotated with various dissonance labels from the humlib filter, including 'P', 'p', 'N', 'n', 's', 'g', 'Z', 'z', 'I', 'i', 'a', and 'v'. These labels are placed below the notes they apply to, often within colored circles or boxes. For example, in the tenor part, there are blue circles around 'P' and 'p', a red circle around 'g', and a green circle around 's'. In the bassus part, there is a purple circle around 'n' and a blue circle around 'P'. The labels are distributed across the score, indicating specific points of dissonance between the voices.

P/p: rising/downward passing tone
N/n: upper/lower neighbor tone
s: suspension
g: agent
Z/z: unclassified dissonance (2nd or 7th / 4th)
I/i: unclassified dissonance in parallel accompaniment

MP Editor with “known” and “unknown” dissonances
(labels come from *humlib’s dissonant filter*,
developed by Alex Morgan and Craig Sapp for the Josquin Research Project)

Interoperability between MP Editor & Dissonance Filter (Humlib)

Craig Sapp, Juliette Regimbal, and Martha Thomae

Does the use of “counterpoint” rules
(an older concept than “harmony”)
help in detecting scribal errors?

Does marking “unknown dissonances” make the correction
of the scoring-up results more efficient?

Yes!



A musical score for a part labeled 'bassus' in common time (C). The staff contains several measures of music with diamond-shaped markers indicating dissonances. Below the staff, there are labels: 'p' (piano) under the first measure, 'P' (piano) and 'g' (guitar) under the last measure, and 's' (soprano) above the staff in the middle. The interface includes a 'Bar by:' dropdown menu set to 'none', a 'Switch to Modern Clef' button, and a red dashed box around the 'Add Dissonance Labels' checkbox, which is currently unchecked. A 'Continue in Editorial Mode' button is also visible.

1. **Marking unknown dissonances** reduces the correction time (at least to a half)
→ allows for a focused search for errors

superius

altus

tenor

bassus

141v.jpg_p1_s1

141v.jpg_p1_s2

141v.jpg_p1_s3

142r.jpg_p1_s1

142r.jpg_p1_s2

142r.jpg_p1_s3

141v.jpg_p1_s5

141v.jpg_p1_s6

141v.jpg_p1_s7

142r.jpg_p1_s6

142r.jpg_p1_s7

142r.jpg_p1_s8

z g

g

g

p

p

s

p l p

w

V

V

W

Z

s

g

s

n

p l v

v

p

s

s

p

g

a

Z

g

P

g

v

v

p

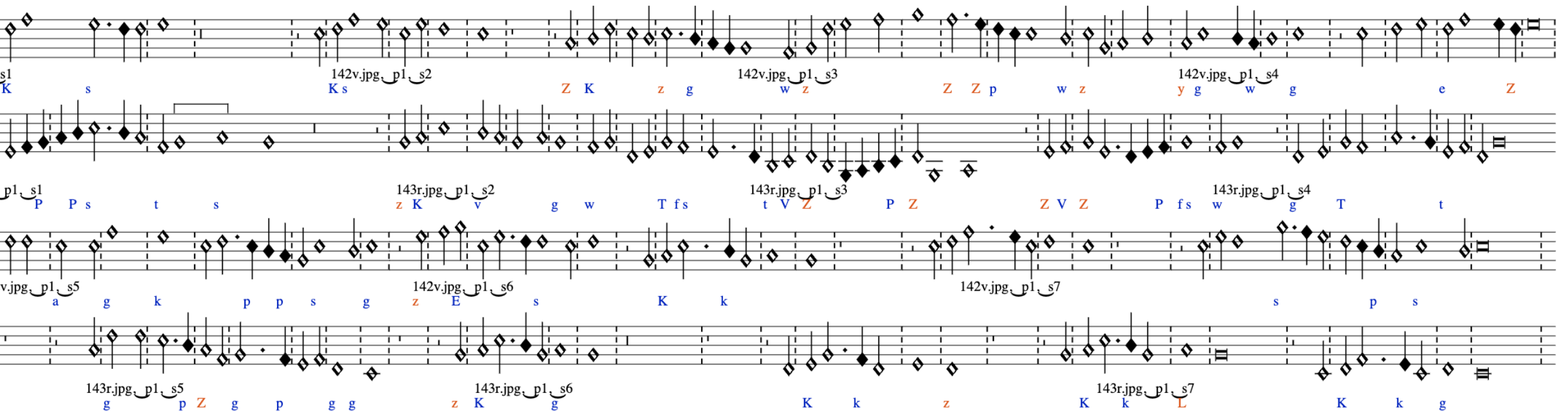
v

v

g

Z

1. **Marking unknown dissonances** reduces the correction time (at least to a half)
→ allows for a focused search for errors



1. **Marking unknown dissonances** reduces the correction time (at least to a half)
→ allows for a focused search for errors

The image displays a musical score for four voices: superius, altus, tenor, and bassus. The score is written in C major and common time (C). Each staff contains a series of notes, some of which are marked with diamond symbols. Below the notes, there are various annotations including file names and segment numbers (e.g., 141v.jpg_p1_s1, 142r.jpg_p1_s1, 141v.jpg_p1_s5, 142r.jpg_p1_s6, 141v.jpg_p1_s7, 142r.jpg_p1_s8) and letters (e.g., z, g, p, s, w, v, n, l, v, p, s, a, g, p, g, v, p, v, v, g, Z). A dashed purple line with a bracket underneath spans across the bottom of the first three staves, indicating a specific region. An orange arrow points to a specific note in the bassus staff, which is labeled with an orange 'Z'.

Region to look at for an error

1st orange
label

2. Marking unknown dissonances increases accuracy of corrections

The image displays a musical score for four voices: superius, altus, tenor, and bassus. The score is written in C major and common time. The notes are represented by diamond shapes. Various annotations are present:

- A red arrow points to a specific note in the altus part.
- Two purple dashed boxes highlight regions in the altus and bassus parts.
- Orange labels 'z' and 'g' are placed above notes in the altus and bassus parts.
- Blue labels 'g', 'p', 's', 'n', 'p', 'l', 'v', 'v', 'p', 'v', 'v', 'g' are placed below notes in the bassus part.
- A purple dashed line with an arrow points to a region in the bassus part.

Below the score, three text boxes provide additional information:

- Region to look at for an error
- 1st orange label
- Cadence does not line up; alto is a minim too long

2. Marking unknown dissonances increases accuracy of corrections

The image displays a musical score for four voices: superius, altus, tenor, and bassus. The score is written in C major and common time (C). The notes are represented by diamond-shaped markers on the staves. Various annotations are present throughout the score, including file names and segment identifiers (e.g., 141v.jpg_p1_s1, 142r.jpg_p1_s1, 141v.jpg_p1_s5, 142r.jpg_p1_s6, 141v.jpg_p1_s6, 142r.jpg_p1_s7, 141v.jpg_p1_s7, 142r.jpg_p1_s8) and letters (e.g., g, p, s, c, f, s, s, n, p, a, g, p, g). A red arrow points to a specific note in the altus part, which is marked with a red 'Z' and 'Z' above it, indicating a dissonance. The score is divided into measures by vertical bar lines.

2. Marking unknown dissonances increases accuracy of corrections

The image displays a musical score for four voices: superius, altus, tenor, and bassus. The score is written in C major, common time (C). The notes are represented by diamond shapes. Various annotations are present throughout the score, including file names and segment numbers (e.g., 141v.jpg_p1_s1, 142r.jpg_p1_s1) and phonetic labels (e.g., z, g, p, l, v, s, n, a, g, P, g, v, p, v, v, g). A red arrow points to a specific note in the altus part. Two purple dashed boxes highlight specific regions: one in the altus part and another in the bassus part. A purple dashed line with a bracket spans across the bottom of the score. Below the score, there are three text boxes:

- Region to look at for an error
- 1st orange label
- Cadence does not line up; alto is a minim too long

2. Marking unknown dissonances increases accuracy of corrections

The image displays a musical score for four voices: superius, altus, tenor, and bassus. The score is written in C major, with a common time signature (C). The notes are represented by diamond shapes. The score is divided into measures by vertical lines. The notes are labeled with letters (g, p, s, c, n, a, P, g) and numbers (141v.jpg_p1_s1, 141v.jpg_p1_s2, 142r.jpg_p1_s1, 142r.jpg_p1_s2, 141v.jpg_p1_s5, 141v.jpg_p1_s6, 141v.jpg_p1_s7, 142r.jpg_p1_s6, 142r.jpg_p1_s7, 142r.jpg_p1_s8). A red arrow points to a dissonance in the altus part, specifically a red diamond shape. A dashed purple box highlights a cadence in the bassus part, specifically a dashed purple diamond shape. A text box at the bottom right says "Cadence lining up" with a dashed purple arrow pointing to the dashed purple box.

superius

altus

tenor

bassus

141v.jpg_p1_s1

141v.jpg_p1_s2

142r.jpg_p1_s1

142r.jpg_p1_s2

141v.jpg_p1_s5

141v.jpg_p1_s6

141v.jpg_p1_s7

142r.jpg_p1_s6

142r.jpg_p1_s7

142r.jpg_p1_s8

Cadence lining up

3. **Marking unknown dissonances** even helped detecting OMR errors that went unnoticed on the previous stage of the pipeline

Digitization & MIR Pipeline

Digitization



Optical music recognition (OMR) with human correction



Automatic scoring up & editorial correction

MEASURING POLYPHONY

DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC



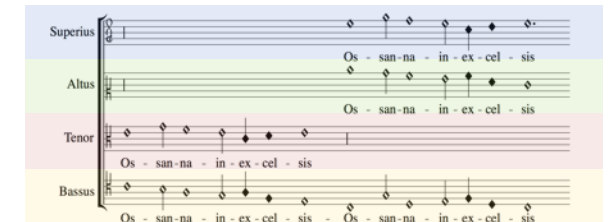
Manuscript



Digital images

```
<mei>
<music>
<staff n="1">
  <clef line="2" shape="G"/>
  <note dur="semibrevis" pname="g" oct="5"/>
  <note dur="semibrevis" pname="d" oct="5"/>
  <dot/>
  <note dur="minima" pname="c" oct="5"/>
  <ligature form="recta">
    <note dur="semibrevis" pname="d" oct="5"/>
    <note dur="semibrevis" pname="g" oct="4"/>
  </ligature>
  <note dur="brevis" pname="b" oct="4"/>
</staff>
</music>
</mei>
```

MEI Parts file
encodes the **symbols** in each
part/voice of the images



MEI Score file
interpreted durations
(i.e., *alteration & imperfection*)
& Humlib's Dissonant
Filter (DF)

Outcomes

Preservation and enhanced access of Guatemala's historical music heritage through:

- Digital images
- Mensural MEI files encoding the separate parts and the scores



Digital images

```
<mei>
<music>
<staff n="1">
<clef line="2" shape="G"/>
<note dur="semibrevis" pname="g" oct="5"/>
<note dur="semibrevis" pname="d" oct="5"/>
<dot/>
<note dur="minima" pname="c" oct="5"/>
<ligature form="recta">
<note dur="semibrevis" pname="d" oct="5"/>
<note dur="semibrevis" pname="g" oct="4"/>
</ligature>
<note dur="brevis" pname="b" oct="4"/>
</staff>
</music>
</mei>
```

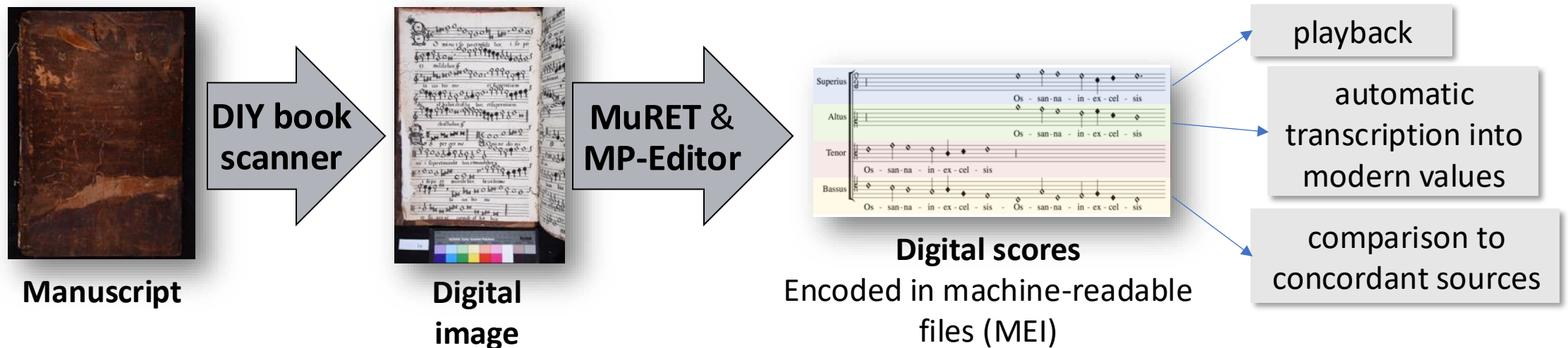
MEI Parts file
encodes the **symbols** in each
part/voice of the images

A musical score for four voices: Superius, Altus, Tenor, and Bassus. Each voice part is on a separate staff. The lyrics are "Os - san - na - in - ex - cel - sis". The notes are diamond-shaped, and the durations are indicated by the shape and position of the notes. The score is color-coded by voice part: Superius (blue), Altus (green), Tenor (pink), and Bassus (yellow).

MEI Score file
interpreted durations
(i.e., *alteration & imperfection*)

Methodology: the digitization and MIR pipeline

- Allows for the **semi-automatic transcription** of mensural music from digital images of manuscripts in mensural notation
- **Applicable** to other mensural music sources
- It considers the use of DIY tools and free, online, and open software to be as accessible as possible to **institutions with limited resources**



Final Remarks

- Following the idea of “preservation and dissemination”
- Focused on using existing software rather than building another tool
- While the main goal of the project was related to preserving cultural heritage (i.e., preserving historical music works)
- Strived to preserve the previous work from people who put the effort and resources into building these MIR tools

Final Remarks

- Idea of “collaboration”
- Collaborating to preserve a heritage that can be lost
- Or collaborating on preserving these MIR technologies by acknowledging their existence and the specialized tasks they were built for, improving them, and bringing them together for a larger task using their individual advantages
- A lot of work can be done by collaborating instead of working in silos

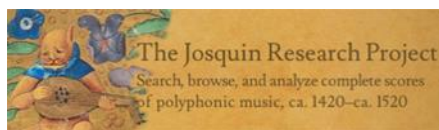
Acknowledgements

This collaborative project would not have been possible without the support from:

- **Archivo Histórico Arquidiocesano de Guatemala (AHAG)** for granting permission for the project
- **Omar Morales Abril**, Guatemalan musicologist that provided me with advice and information about the choirbooks
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- **José German Thomae Villela** for constructing the cradle
- **Darryl Cameron** for lending the copystand
- **Daniel Hernández-Salazar** for setting the camera parameters
- **Distributed Digital Music Archives and Libraries (DDMAL) Lab at McGill University** for lending the lights and other parts of the equipment
- **Marvin Duchow Music Library at McGill University** for lending the camera
- **Centro de Rescate, Estudio y Análisis Científico del Arte (CREA)** for the conservation assessment of the collection and the conservation treatment of the choirbook

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- **David Rizo**, **Karen Desmond**, and **Juliette Regimbal** who made possible for MuRET and the MP-Editor to communicate with one another
- **Craig Stuart Sapp** whose work allowed for the dissonance labels to be displayed in the MP-Editor
- **Jorge Calvo-Zaragoza**, **Francisco Castellanos**, **Antonio Ríos Vila**, and **José Manuel Iñesta** who developed the models used in MuRET
- **Laurent Pugin** for his work on Verovio
- **Alexander Morgan** for his Renaissance dissonance labeler script
- **Peter Schubert** for his expert advice in counterpoint matters
- **Geneviève Gates-Panneton** for her work in the experiment
- My **supervisors**: **Julie Cumming** and **Ichiro Fujinaga**



Social Sciences and Humanities
Research Council of Canada

Conseil de recherches en
sciences humaines du Canada

Canada



Thank you!

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<https://martha-thomae.github.io/projects/guatemala.html>